

# INVERSE TRIGONOMETRIC FUNCTIONS

## WORKED OUT EXAMPLES

**W.E-1:** Show that

$$3 \tan^{-1}\left(\frac{1}{2}\right) + 2 \tan^{-1}\left(\frac{1}{5}\right) + \sin^{-1}\left(\frac{142}{65\sqrt{5}}\right) = \pi$$

Sol:

$$3 \tan^{-1}\left(\frac{1}{2}\right) = \tan^{-1}\left(\frac{\frac{3}{2} - \frac{1}{8}}{1 - 3 \times \frac{1}{4}}\right) = \tan^{-1}\left(\frac{11}{2}\right)$$

$$2 \tan^{-1}\left(\frac{1}{5}\right) = \tan^{-1}\left(\frac{\frac{2}{5}}{1 - \frac{1}{25}}\right) = \tan^{-1}\left(\frac{5}{12}\right)$$

$$\therefore L.H.S = \pi + \tan^{-1}\left(\frac{\frac{11}{2} + \frac{5}{12}}{1 - \frac{11}{2} \cdot \frac{5}{12}}\right) + \sin^{-1}\left(\frac{142}{65\sqrt{5}}\right)$$

$$\left(\because \frac{11}{2} \times \frac{5}{12} > 1\right)$$

$$= \pi - \tan^{-1}\left(\frac{71}{12} \times \frac{24}{31}\right) + \sin^{-1}\left(\frac{142}{65\sqrt{5}}\right)$$

$$= \pi - \tan^{-1}\left(\frac{142}{31}\right) + \tan^{-1}\left(\frac{142}{31}\right) = \pi$$

**W.E-2:**

$$\text{Let } y = \sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7).$$

If  $y$  simplifies to  $a\pi + b$  then find  $a - b$

Sol:

$$y = 3\pi - 8 - (10 - 3\pi) + 4\pi - 12 - (9 - 2\pi)$$

$$+ 6 - \pi - (7 - 2\pi)$$

$$\Rightarrow y = 13\pi - 40$$

$$\therefore a - b = 53$$

**W.E-3:** Prove that

$$\sin\left(\cot^{-1}\left(\tan\left(\cos^{-1}x\right)\right)\right) = \sin\left(\operatorname{cosec}^{-1}\cot\left(\tan^{-1}x\right)\right) = x$$

$$\text{Where } x \in (0, 1]$$

$$\text{Sol: Let } \cos^{-1}x = \alpha \Rightarrow x = \cos \alpha \Rightarrow \tan \alpha = \frac{\sqrt{1-x^2}}{x}$$

$$\therefore \sin\left(\cot^{-1}\left(\tan\left(\cos^{-1}x\right)\right)\right)$$

$$= \sin\left(\cot^{-1}\left(\cot\left(\frac{\pi}{2} - \alpha\right)\right)\right) = \cos \alpha = x$$

$$\text{Let } \tan^{-1}x = \beta \Rightarrow x = \tan \beta \Rightarrow \cot \beta = \frac{1}{x}$$

$$\therefore \sin\left(\operatorname{cosec}^{-1}\left(\frac{1}{x}\right)\right) = \sin\left(\sin^{-1}x\right) = x$$

**W.E-4:**

If  $a, b, c$  have the same sign then find the value of

$$\cot^{-1}\left(\frac{ab+1}{a-b}\right) + \cot^{-1}\left(\frac{bc+1}{b-c}\right) + \cot^{-1}\left(\frac{ca+1}{c-a}\right)$$

Sol: case (i): Let  $a > b > c \Rightarrow a - b > 0$ ,

$$b - c > 0 \text{ but } c - a < 0$$

$\therefore$  Required value

$$= \tan^{-1}a - \tan^{-1}b + \tan^{-1}b - \tan^{-1}c + \pi + \tan^{-1}c - \tan^{-1}a$$

$$= \pi$$

case (ii): Let  $a < b < c \Rightarrow a - b < 0, b - c < 0, c - a > 0$

$\therefore$  Required value

$$= \pi + \tan^{-1}\left(\frac{a-b}{1+ab}\right) + \pi + \tan^{-1}\left(\frac{b-c}{1+bc}\right) + \tan^{-1}\left(\frac{c-a}{1+ca}\right)$$

$$= \pi + \tan^{-1} a - \tan^{-1} b + \pi + \tan^{-1} b - \tan^{-1} c + \tan^{-1} c - \tan^{-1} a$$

$$= 2\pi$$

**W.E -5:** Let  $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$  and

$f: R \rightarrow \left(0, \frac{\pi}{2}\right]$  then find the complete set of real values of  $\alpha$  for which  $f(x)$  is onto.

Sol:  $x^2 + 4x + \alpha^2 - \alpha \geq 0 \Rightarrow 16 - 4(\alpha^2 - \alpha) \leq 0$

$$\Rightarrow \alpha^2 - \alpha - 4 \geq 0$$

but  $\cot^{-1}(x^2 + 4x + \alpha^2 - \alpha) = \frac{\pi}{2}$

$$\Rightarrow x^2 + 4x + \alpha^2 - \alpha = 0 \text{ and } x \in R$$

$$\Rightarrow 16 - 4(\alpha^2 - \alpha) \geq 0 \Rightarrow \alpha^2 - \alpha - 4 \leq 0$$

$$\therefore \alpha^2 - \alpha - 4 = 0 \Rightarrow \alpha = \frac{1 \pm \sqrt{17}}{2}$$

**W.E -6:** Solve:  $\cos^{-1} \frac{x^2 - 1}{x^2 + 1} + \tan^{-1} \frac{2x}{x^2 - 1} = \frac{2\pi}{3}$

Sol: case (i): Let  $0 < x < 1$

$\therefore$  given equation is

$$\pi - \cos^{-1} \left( \frac{1 - x^2}{1 + x^2} \right) - \tan^{-1} \left( \frac{2x}{1 - x^2} \right) = \frac{2\pi}{3}$$

$$\Rightarrow \frac{\pi}{3} = 4 \tan^{-1} x \Rightarrow x = \tan \left( \frac{\pi}{12} \right) = 2 - \sqrt{3}$$

case (ii): Let  $x > 1$   $\therefore$  given equation is

$$\pi - 2 \tan^{-1} x - (-\pi + 2 \tan^{-1} x) = \frac{2\pi}{3}$$

$$\left( \therefore \tan^{-1} \frac{2x}{1 - x^2} = \pi + 2 \tan^{-1} x \text{ if } x > 1, \cos^{-1} \frac{1 - x^2}{1 + x^2} = 2 \tan^{-1} x \text{ if } x \geq 0 \right)$$

$$\Rightarrow \frac{4\pi}{3} = 4 \tan^{-1} x \Rightarrow x = \sqrt{3}$$

Case (iii) Let  $-1 < x < 0$

$\therefore$  given equation is

$$\pi - (-2 \tan^{-1} x) - 2 \tan^{-1} x = \frac{2\pi}{3}$$

$$\Rightarrow \pi = \frac{2\pi}{3} \text{ absurd}$$

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Case (iv) Let  $x < -1$

$\therefore$  given equation is

$$\pi - (-2 \tan^{-1} x) - (\pi + 2 \tan^{-1} x) = \frac{2\pi}{3}$$

$$\Rightarrow 0 = \frac{2\pi}{3} \text{ absurd}$$

$$\therefore x = 2 - \sqrt{3} \text{ or } \sqrt{3} \text{ are the solutions}$$

**W.E -7:** If  $x \in \left[-1, -\frac{1}{2}\right]$  then express the function

$$f(x) = \sin^{-1}(3x - 4x^3) + \cos^{-1}(4x^3 - 3x) \text{ in}$$

the form of  $a \cos^{-1} x + b\pi$  where  $a$  and  $b$  are rational numbers.

Sol: Let  $x = \cos \theta \Rightarrow \frac{2\pi}{3} \leq \theta \leq \pi \Rightarrow 2\pi \leq 3\theta \leq 3\pi$

$$\Rightarrow 0 \leq 3\theta - 2\pi \leq \pi$$

$$\therefore f(x) = \sin^{-1}(-\cos 3\theta) + \cos^{-1}(\cos 3\theta)$$

$$= -\sin^{-1}(\cos 3\theta) + \cos^{-1}(\cos(3\theta - 2\pi))$$

$$(\because -3\pi \leq -3\theta \leq -2\pi)$$

$$= -\sin^{-1} \left( \sin \left( \frac{5\pi}{2} - 3\theta \right) \right) + 3\theta - 2\pi$$

$$= -\frac{5\pi}{2} + 3\theta + 3\theta - 2\pi$$

$$\left( \because -\frac{\pi}{2} \leq \frac{5\pi}{2} - 3\theta \leq \frac{\pi}{2} \right)$$

$$= 6\theta - \frac{9\pi}{2} = 6 \cos^{-1} x - \frac{9\pi}{2}$$

**W.E -8:** Find all real values of  $K$  for which there is a triangle whose angles have measure

$$\tan^{-1} \left( \frac{1}{2} \right), \tan^{-1} \left( \frac{1}{2} + K \right) \text{ and } \tan^{-1} \left( \frac{1}{2} + 2K \right)$$

Sol: we have

$$\tan^{-1} \left( \frac{1}{2} \right) + \tan^{-1} \left( \frac{1}{2} + K \right) + \tan^{-1} \left( \frac{1}{2} + 2K \right) = \pi$$

$$\Rightarrow \tan^{-1} \left( \frac{1}{2} + K \right) + \tan^{-1} \left( \frac{1}{2} + 2K \right) = \pi - \tan^{-1} \left( \frac{1}{2} \right)$$

$$\Rightarrow \frac{\frac{1}{2} + K + \frac{1}{2} + 2K}{1 - \left(\frac{1}{2} + K\right)\left(\frac{1}{2} + 2K\right)} = -\frac{1}{2}$$

$$\Rightarrow 8K^2 - 18K - 11 = 0 \Rightarrow K = -\frac{1}{2} \text{ or } K = \frac{11}{4}$$

but for  $K = -\frac{1}{2}$  they can't be the angles of a triangle

$$\therefore K = \frac{11}{4}$$

**W.E -9:** If the sum  $\sum_{n=1}^{10} \sum_{m=1}^{10} \tan^{-1}\left(\frac{m}{n}\right) = K\pi$  then the value of K is

Sol:  $\tan^{-1}\left(\frac{1}{1}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \dots + \tan^{-1}\left(\frac{1}{10}\right)$   
 $+ \tan^{-1}\left(\frac{2}{1}\right) + \tan^{-1}\left(\frac{2}{2}\right) + \dots + \tan^{-1}\left(\frac{2}{10}\right)$   
 $+ \tan^{-1}\left(\frac{3}{1}\right) + \tan^{-1}\left(\frac{3}{2}\right) + \dots + \tan^{-1}\left(\frac{3}{10}\right) + \dots +$   
 $\tan^{-1}\left(\frac{10}{1}\right) + \tan^{-1}\left(\frac{10}{2}\right) + \tan^{-1}\left(\frac{10}{3}\right) + \dots + \tan^{-1}\left(\frac{10}{10}\right)$   
 $= 10 \times \frac{\pi}{4} + 45 \times \frac{\pi}{2} \left( \because \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2} \right)$   
 $= \frac{5\pi}{2} + \frac{45\pi}{2} = \frac{50\pi}{2} = 25\pi \Rightarrow K = 25$

**W.E -10:** If the value of

$$\lim_{n \rightarrow \infty} \sum_{K=2}^n \cos^{-1} \left( \frac{1 + \sqrt{(k-1)K(K+1)(K+2)}}{K(K+1)} \right)$$

is equal to  $\frac{\pi}{K}$  then the value of K is

Sol:

$$\sum_{k=2}^n \cos^{-1} \left( \frac{1}{k(k+1)} + \sqrt{\frac{(k-1)(k+2)}{k(k+1)}} \right)$$

$$= \sum_{K=2}^n \cos^{-1} \left( \frac{1}{K(K+1)} + \sqrt{\left(1 - \frac{1}{K^2}\right)\left(1 - \frac{1}{(K+1)^2}\right)} \right)$$

$$= \sum_{K=2}^n \left( \cos^{-1} \frac{1}{K+1} - \cos^{-1} \frac{1}{K} \right)$$

$$= \cos^{-1}\left(\frac{1}{3}\right) - \cos^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{4}\right) - \cos^{-1}\left(\frac{1}{3}\right) + \dots + \cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{n}\right)$$

$$= \lim_{n \rightarrow \infty} \left( \cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{2}\right) \right)$$

$$= \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6} \quad \therefore K = 6$$

**W.E-11:** The value of

$$\cot \left( \sum_{n=1}^{23} \cot^{-1} \left( 1 + \sum_{k=1}^n 2k \right) \right) \text{ is}$$

(ADV-2013)

A)  $\frac{23}{25}$     B)  $\frac{25}{23}$     C)  $\frac{23}{24}$     D)  $\frac{24}{23}$

Sol: (B)  $\cot \left( \sum_{n=1}^{23} \cot^{-1} (n^2 + n + 1) \right)$

$$\cot \left( \sum_{n=1}^{23} \tan^{-1} \left( \frac{n+1-n}{1+n(n+1)} \right) \right)$$

$$\Rightarrow \cot \left( \tan^{-1} \left( \frac{23}{25} \right) \right) = \frac{25}{23}$$

**W.E-12:** Let  $f : [0, 4\pi] \rightarrow [0, \pi]$  be defined

by  $f(x) = \cos^{-1}(\cos x)$ . The number of points  $x \in [0, 4\pi]$  satisfying the equation

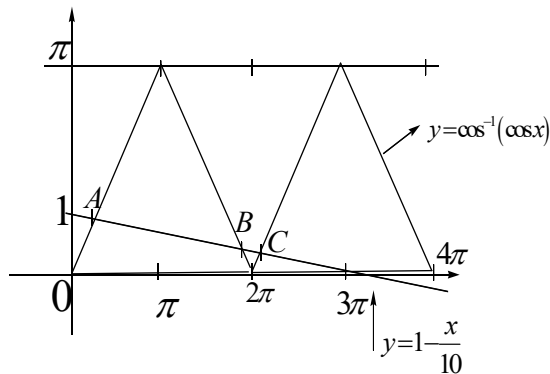
$$f(x) = \frac{10-x}{10} \text{ is} \quad \text{(ADV-2014)}$$

Sol: (3)

$$f : [0, 4\pi] \rightarrow [0, \pi], f(x) = \cos^{-1}(\cos x)$$

$$\Rightarrow \text{points A, B, C satisfy } f(x) = \frac{10-x}{10}$$

Hence, 3 points



### LEVEL-V

#### SINGLE ANSWER TYPE QUESTIONS

1. If we consider only the principle values of the inverse trigonometric functions then the

value of  $\tan\left(\cos^{-1}\frac{1}{5\sqrt{2}} - \sin^{-1}\frac{4}{\sqrt{17}}\right)$  is  
(IIT 1994)

- (A)  $\frac{\sqrt{29}}{3}$  (B)  $\frac{29}{3}$  (C)  $\frac{\sqrt{3}}{29}$  (D)  $\frac{3}{29}$

2. The product of all values of  $x$  satisfying the equation

$$\sin^{-1} \cos\left(\frac{2x^2 + 10|x| + 4}{x^2 + 5|x| + 3}\right) = \cot\left(\cot^{-1}\left(\frac{2 - 18|x|}{9|x|}\right)\right) + \frac{\pi}{2}$$

- (A) -1 (B) +9 (C) -9 (D) 3

3. Given  $0 \leq x \leq \frac{1}{2}$  then the value of

$$\tan\left[\sin^{-1}\left\{\frac{x}{\sqrt{2}} + \frac{\sqrt{1-x^2}}{\sqrt{2}}\right\} - \sin^{-1}x\right] \text{ is}$$

- (A) -1 (B) 1 (C)  $\frac{1}{\sqrt{3}}$  (D)  $\sqrt{3}$

4. If  $0 < x < 1$ , then

$$\sqrt{1+x^2} \left[ \left\{ x \cos(\cot^{-1} x) + \sin(\cot^{-1} x) \right\}^2 - 1 \right]^{\frac{1}{2}}$$

is equal to

[IIT-2008]

#### INVERSE TRIGONOMETRIC FUNCTIONS

- (A)  $\frac{x}{\sqrt{1+x^2}}$  (B)  $x$   
(C)  $x\sqrt{1+x^2}$  (D)  $\sqrt{1+x^2}$

5. If  $(\sin^{-1}a)^2 + (\cos^{-1}b)^2 + (\sec^{-1}c)^2 + (\operatorname{cosec}^{-1}d)^2 = \frac{5\pi^2}{2}$ , then the value of

$(\sin^{-1}a)^2 - (\cos^{-1}b)^2 + (\sec^{-1}c)^2 - (\operatorname{cosec}^{-1}d)^2$  is equal to

- (A)  $-\pi^2$  (B)  $-\frac{\pi^2}{2}$  (C) 0 (D)  $\frac{\pi^2}{2}$

6. If  $a \sin^{-1}x - b \cos^{-1}x = c$ , then the value of  $a \sin^{-1}x + b \cos^{-1}x$  (whenever exists) is equal to

- (A) 0 (B)  $\frac{\pi ab + c(b-a)}{a+b}$   
(C)  $\frac{\pi}{2}$  (D)  $\frac{\pi ab + c(a-b)}{a+b}$

7. Sum of infinite terms of the series

$$\cot^{-1}\left(1^2 + \frac{3}{4}\right) + \cot^{-1}\left(2^2 + \frac{3}{4}\right) + \cot^{-1}\left(3^2 + \frac{3}{4}\right) + \dots$$

is

- (A)  $\pi/4$  (B)  $\tan^{-1}2$   
(C)  $\tan^{-1}3$  (D)  $\tan^{-1}4$

8. If  $0 < \cos^{-1}x < 1$  and

$$1 + \sin(\cos^{-1}x) + \sin^2(\cos^{-1}x)$$

$$+ \sin^3(\cos^{-1}x) + \dots = 2, \text{ then}$$

the value of  $12x^2$  is

- (A) 1 (B) 4 (C) 9 (D) 0

9. If  $\log_2 x \geq 0$  then

$$\log_{\frac{1}{\pi}}\left(\sin^{-1}\left(\frac{2x}{1+x^2}\right) + 2\tan^{-1}x\right) = \lambda$$

then  $|\lambda| =$

- (A) 1 (B) 2 (C) 3 (D) 4

10. For the equation

$$2x = \tan(2 \tan^{-1}a) + 2 \tan(\tan^{-1}a + \tan^{-1}a^3),$$

which of the following is valid?

- (A)  $a^2x + 2a = x$  (B)  $a^2 + 2ax + 1 = 0$

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JEE ADVANCED - VOL - II

- (C)  $a \neq 0$  (D)  $a = -2, 2$
11. Number of solutions of the equation  $\cos^{-1}(1-x) - 2\cos^{-1}x = \frac{\pi}{2}$  is  
 A) 3 B) 2 C) 1 D) 0
12. If  $\cos^{-1}x = \tan^{-1}x$ , then  
 (A)  $x^2 = \frac{\sqrt{5}-1}{2}$  (B)  $x^2 = \frac{\sqrt{5}+1}{2}$   
 (C)  $\sin(\cos^{-1}x) = \frac{\sqrt{3}-1}{2}$   
 (D)  $\tan(\cos^{-1}x) = \frac{\sqrt{5}-1}{2}$
13. The number of solutions for the equation  $\cos^{-1}(1-x) + m\cos^{-1}x = \frac{n\pi}{2}$  where  $m > 0, n \leq 0$  is  
 (A) 0 (B) 1 (C) 2 (D) Infinite
14. The solution set of the equation  $\sin^{-1}\sqrt{1-x^2} + \cos^{-1}x = \cot^{-1}\frac{\sqrt{1-x^2}}{x} - \sin^{-1}x$  is  
 (A)  $[-1, 1] - \{0\}$  (B)  $[-1, 0) \cup \{1\}$   
 (C)  $(0, 1] \cup \{-1\}$  (D)  $[-1, 1]$
15. The sum of the solutions of the equation  $2\sin^{-1}\sqrt{x^2+x+1} + \cos^{-1}\sqrt{x^2+x} = \frac{3\pi}{2}$  is  
 A) 0 B) -1 C) 3 D) 2
16. The number of integral values of  $k$  for which the equation  $\sin^{-1}x + \tan^{-1}x = 2k + 1$  has a solution is  
 A) 1 B) 2 C) 3 D) 4
17. The value of  $x$  for which  $\sin[\cot^{-1}(1+x)] = \cos(\tan^{-1}x)$  is : (IIT 2004)  
 (A)  $\frac{1}{2}$  (B) 1 (C) 0 (D)  $-\frac{1}{2}$
18. If  $\tan^{-1}(\sin^2\theta - 2\sin\theta + 3)$

$+ \cot^{-1}(5^{\sec^2\beta} + 1) = \frac{\pi}{2}$ , then the value of

$\cos^2\theta - \sin\theta$  is  
 A) 0 B) -1 C) 1 D) 2

19. If  $\sin^{-1}(\cos^{-1}x) < 1$  and  $\cos^{-1}(\cos^{-1}x) < 1$  then  $x \in$   
 A)  $(\sin 1, \sin(\sin 1))$   
 B)  $(\cos(\cos 1), \cos(\sin 1))$   
 C)  $(\cos(\sin 1), \cos(\cos 1))$   
 D)  $\phi$  (empty set)
20. The set of values of  $\lambda$  for which  $x^2 - \lambda x + \sin^{-1}(\sin 4) > 0$  for all  $x \in \mathbb{R}$ , is  
 (A)  $\phi$  (B)  $(-2, 2)$  (C)  $\mathbb{R}$  (D)  $(-3, 3)$
21. The solution of the inequality  $(\sec^{-1}x)^2 - 6\sec^{-1}x + 8 > 0$  is  
 (A)  $(-\infty, \sec 2) \cup [1, \infty)$  (B)  $(\sec 2, 1)$   
 (C)  $[\sec 2, 1]$  (D)  $(-\infty, \sec 2] \cup [1, \infty)$
22. Number of common points for the curves  $y = \sin^{-1}(2x) + \tan^{-1}\left(\frac{1}{[2x]}\right) + 2$  and  $y = \cos^{-1}(2x+5) + 1$  is (where  $[.]$  denotes greatest integer function)  
 (A) 0 (B) 1 (C) 3 (D) 4
23. The value of  $\sin^{-1}(\cos 2) - \cos^{-1}(\sin 2) + \tan^{-1}(\cot 4) - \cot^{-1}(\tan 4) + \sec^{-1}(\csc 6) - \csc^{-1}(\sec 6)$  is  
 (A) 0 (B)  $3\pi$  (C)  $8 - 3\pi$  (D)  $5\pi - 16$
24. The number of real solutions of  $|y| = \sin x, y = \cos^{-1}(\cos x), -2\pi \leq x \leq 2\pi$  is  
 (A) 2 (B) 1 (C) 3 (D) 4
25. Let  $f(x) = \frac{2}{\pi}(\sin^{-1}[x] + \tan^{-1}[x] + \cot^{-1}[x])$  where  $[x]$  denotes greatest integer less than or equal to  $x$ . If A and B denote the domain and range of  $f(x)$  respectively, then the number of integers in  $(A \cup B)$ , is  
 (A) 1 (B) 2 (C) 3 (D) 4

## MULTIPLE ANSWER TYPE QUESTIONS

26. If  $(\sin^{-1} x)^2 + (\sin^{-1} y)^2 + (\sin^{-1} z)^2 = \frac{3\pi^2}{4}$ ,

then the value of  $(x - y + z)$  can be

- (A) 1 (B) -1 (C) 3 (D) -3

27. Which of the following is/are true

(A)  $2\tan^{-1} x = \sin^{-1} \left( \frac{2x}{1+x^2} \right)$ , when  $|x| \leq 1$

(B)  $2 \sin^{-1} x = \sin^{-1} \{ [2x\sqrt{1-x^2}] \}$ , when

$$-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$$

(C)  $2\tan^{-1} x = \sin^{-1} \left( \frac{2x}{1+x^2} \right)$ , when  $x > 1$

(D)  $2\tan^{-1} x = \sin^{-1} \left( \frac{2x}{1+x^2} \right)$ , when  $x < -1$

28. Which of the following is/are true

A)  $\tan^{-1} \frac{1}{3} = \frac{1}{2} \sin^{-1} \frac{3}{5}$

B)  $\tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \cot^{-1} 2$

C)  $\tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \frac{1}{2} \sin^{-1} \frac{4}{5}$

D)  $\tan^{-1} \frac{1}{3} = \frac{\pi}{2} - \cot^{-1} 3$

29. Which of the following is a rational number?

A)  $\sin \left( \tan^{-1} 3 + \tan^{-1} \frac{1}{3} \right)$

B)  $\cos \left( \frac{\pi}{2} - \sin^{-1} \frac{3}{4} \right)$

C)  $\log_2 \left( \sin \left( \frac{1}{4} \sin^{-1} \frac{\sqrt{63}}{8} \right) \right)$

D)  $\tan \left( \frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3} \right)$

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30.  $\theta = \tan^{-1}(2 \tan^2 \theta) - \tan^{-1} \left( \frac{1}{3} \tan \theta \right)$  if

- (A)  $\tan \theta = -2$  (B)  $\tan \theta = 0$   
(C)  $\tan \theta = 1$  (D)  $\tan \theta = 2$

31. Let  $f(x) = e^{\cos^{-1} \sin(x+\pi/3)}$  then

(A)  $f\left(\frac{8\pi}{9}\right) = e^{5\pi/18}$  (B)  $f\left(\frac{8\pi}{9}\right) = e^{13\pi/18}$

(C)  $f\left(-\frac{7\pi}{4}\right) = e^{\pi/12}$  (D)  $f\left(-\frac{7\pi}{4}\right) = e^{11\pi/12}$

32. Let  $f(x) = \sin^{-1} x + \cos^{-1} x$ . Then  $\pi/2$  is equal to

(A)  $f\left(-\frac{1}{2}\right)$  (B)  $f(k^2 - 2k + 3)$ ,  $k \in \mathbb{R}$

(C)  $f\left(\frac{1}{1+k^2}\right)$ ,  $k \in \mathbb{R}$  (D)  $f(-2)$

33. The values of  $x$  satisfying  $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$  is/are

- (A) 0 (B)  $\frac{1}{2}$  (C) 1 (D) 2

34. If  $\alpha, \beta$  and  $\gamma$  are the roots of  $\tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1} 3x$ , then

A)  $\alpha + \beta + \gamma = 0$  B)  $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{-1}{4}$

C)  $\alpha\beta\gamma = 1$  D)  $|\alpha - \beta|_{\max} = 1$

35. Number of integral solutions of the equation

$\sin^{-1}(\sin x) = \cos^{-1}(\cos x)$  in  $[0, 5\pi]$  is

- A) 2 B) 3 C) 4 D) infinite

36. If  $\tan^{-1} \left( \frac{x-1}{x-2} \right) + \tan^{-1} \left( \frac{x+1}{x+2} \right) = \frac{\pi}{4}$ , then  $x$  is

- (A)  $-\frac{1}{\sqrt{2}}$  (B) 0 (C)  $\frac{1}{\sqrt{2}}$  (D)  $\frac{\sqrt{3}}{2}$

37. The value(s) of  $x$  satisfying the equation

$\sin^{-1} |\sin x| = \sqrt{\sin^{-1} |\sin x|}$  is/are given by

( $n$  is any integer)

- (A)  $n\pi - 1$  (B)  $n\pi$

- (C)  $n\pi + 1$  (D)  $(2n+1)\frac{\pi}{2} + 1$
38.  $\sin^{-1}(x^2 + 2x + 2) + \tan^{-1}(x^2 - 3x - k^2) > \frac{\pi}{2}$   
for  $k \in$   
(A)  $(-1, 0)$  (B)  $(0, 1)$  (C)  $(1, 2)$  (D)  $(0, 2)$
39. Which of the following quantities is/are positive?  
A)  $\cos(\tan^{-1}(\tan 4))$  B)  $\sin(\cot^{-1}(\cot 4))$   
C)  $\tan(\cos^{-1}(\cos 5))$  D)  $\cot(\sin^{-1}(\sin 4))$
40. Indicate the relation which is true  
(A)  $\tan|\tan^{-1} x| = |x|$  (B)  $\cot|\cot^{-1} x| = x$   
(C)  $\tan^{-1}|\tan x| = |x|$  (D)  $\sin|\sin^{-1} x| = |x|$
41. Which of the following are incorrect  
(A)  $\left[\min_{x \in \mathbb{R}} e^x\right] \neq \left[e^{-\pi/2}\right]$ , ( $[.]$  = greatest integer function)  
(B)  $[\sin(\sin x)] > [\cos(\cos x)]$ , where  
 $x \in \left(0, \frac{\pi}{2}\right)$ , ( $[.]$  = Greatest integer function)  
(C)  $\sin 1 > \cos(\sin 1)$   
(D)  $\cos^{-1}\left(\frac{1}{4}\right) < \sin^{-1}\left(\frac{1}{4}\right)$

### COMPREHENSION

This section contains 2 paragraphs. Based upon each paragraph, with multiple choice questions have to be answered. Each question has 4 choices (A), (B), (C) and (D) out of which ONLY ONE is correct.

**Passage-1:**

Let  $\cos^{-1}(4x^3 - 3x) = a + b \cos^{-1} x$

42. If  $x \in \left[-1, -\frac{1}{2}\right)$ , then the value of  $a + b\pi$  is  
A)  $2\pi$  B)  $3\pi$  C)  $\pi$  D)  $-2\pi$

43. If  $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ , then the principal value of  $\sin^{-1}\left(\sin \frac{a}{b}\right)$  is  
A)  $-\frac{\pi}{3}$  B)  $\frac{\pi}{3}$  C)  $-\frac{\pi}{6}$  D)  $\frac{\pi}{6}$
44. If  $x \in \left[\frac{1}{2}, 1\right]$ , then the value of  $\lim_{y \rightarrow a} b \cos(y)$  is  
A)  $-1/3$  B)  $-3$  C)  $1/3$  D)  $3$

**Passage-2:** Consider the system of equations

$$\cos^{-1} x + (\sin^{-1} y)^2 = \frac{p\pi^2}{4} \text{ and}$$

$$(\cos^{-1} x)(\sin^{-1} y)^2 = \frac{\pi^4}{16}, p \in \mathbb{Z}$$

45. The value of  $p$  for which system has a solution is  
A) 1 B) 2 C) 0 D) -1
46. The value of  $x$  which satisfies the system of equation is  
A)  $\cos \frac{\pi^2}{8}$  B)  $\sin \frac{\pi^2}{4}$   
C)  $\cos \frac{\pi^2}{2}$  D)  $\cos \frac{\pi^2}{4}$
47. Which of the following is not the value of  $y$  that satisfies the system of equations  
A) 1 B) -1  
C)  $1/2$  D) All of these

### MATRIX MATCHING QUESTIONS

48. Match the items of Column-I with the items Column-II  
**Column - I**

(A) If  $\cos^{-1} \lambda + \cos^{-1} \mu + \cos^{-1} \gamma = 3\pi$  then  $\lambda\mu + \mu\gamma + \gamma\lambda$  is

(B) If  $\sum_{i=1}^{10} \cos^{-1} x = 0$  then  $\sum_{i=1}^{10} x_i$  is

(C) If  $\sum_{i=1}^{2n} \sin^{-1} x_i = n\pi$  then  $\sum_{i=1}^{2n} x_i$  is

(D) The value of  $\sin^{-1}\left\{\frac{\sqrt{3}}{2}x - \frac{1}{2}\sqrt{1-x^2}\right\}$ ,

$\left(-\frac{1}{2} \leq x \leq \frac{1}{2}\right)$  is

**Column-II**

(P)  $2n$

(Q)  $\left(\sin^{-1}x - \frac{\pi}{6}\right)$

(R) 10

(S) 3

49. **Column-I**

**Column-II**

(A)  $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right]$

(P)  $\frac{3-\sqrt{5}}{2}$

(B)  $\cos\left[\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \frac{\pi}{6}\right]$

(Q)  $\frac{-24}{7}$

(C)  $\tan\left[\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right]$

(R) 1

(D)  $\tan\left[2\sin^{-1}\frac{4}{5}\right]$

(S) -1

### STATEMENT TYPE QUESTIONS

The following questions consist of two statements, one labelled as 'Statement I' and the other 'Statement II'. You have to examine these two statements carefully and decide if the Statement I and the Statement II, are individually true and if so, whether the Statement II is the correct explanation for the given Statement I. Select your answer to these items using the codes given below and then select the correct option.

(A) Both I and II are individually true and II is the correct explanation of I

(B) Both I and II are individually true but II is not the correct explanation of I

(C) I is true but II is false

(D) I is false but II is true

50. **Statement-1:**

$$\operatorname{cosec}^{-1}\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right) > \sec^{-1}\left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)$$

**Statement 2:**  $\operatorname{cosec}^{-1}x < \sec^{-1}x$  if  $1 \leq x < \sqrt{2}$

### INVERSE TRIGONOMETRIC FUNCTIONS

51. **Statement-1:** The value of

$$\tan^{-1}2 + \tan^{-1}3 = \frac{3\pi}{4}$$

**Statement-2:** If  $x > 0, y > 0, xy > 1$  then

$$\tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

52. **Statement-1:**  $\sin^{-1}\left(\frac{1}{\sqrt{e}}\right) > \tan^{-1}\left(\frac{1}{\sqrt{\pi}}\right)$

**Statement-2:**

$$\sin^{-1}x > \tan^{-1}y \text{ for } x > y, \forall x, y \in (0,1)$$

53. **Statement-1:** If  $x < 0, \tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}$

**Statement-2:**  $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}, x \in \mathbb{R}$

### INTEGER TYPE QUESTIONS

54. The solution set of inequality

$$(\cot^{-1}x)(\tan^{-1}x) + \left(2 - \frac{\pi}{2}\right)\cot^{-1}x$$

$$-3\tan^{-1}x - 3\left(2 - \frac{\pi}{2}\right) > 0 \text{ is (a,b) then the}$$

value of  $\cot^{-1}a + \cot^{-1}b$  is

55. If  $x = \sin^{-1}(a^6 + 1) + \cos^{-1}(a^4 + 1)$

$-\tan^{-1}(a^2 + 1), a \in \mathbb{R}$  then the value of  $\sec^2 x$  is

56. If  $0 < \cos^{-1}x < 1$  and  $1 + \sin(\cos^{-1}x)$

$$+ \sin^2(\cos^{-1}x) + \sin^3(\cos^{-1}x) + \dots = 2$$

then the value of  $12x^2$  is

57. If range of the function

$$f(x) = \sin^{-1}x + 2\tan^{-1}x + x^2 + 4x + 1 \text{ is}$$

$[p, q]$  then the value of  $(p+q)$  is

58. If  $\mathbf{r} = \mathbf{x} + \mathbf{y} + \mathbf{z}$  and

$$\tan^{-1}\sqrt{\frac{xr}{yz}} + \tan^{-1}\sqrt{\frac{yr}{zx}} + \tan^{-1}\sqrt{\frac{zr}{xy}} = k\pi$$

then the value of  $k$  is

59. Absolute value of sum of all integral values of  $x$  in the domain of



## INVERSE TRIGONOMETRIC FUNCTIONS

## JEE ADVANCED - VOL - II

$f(x) = \cot^{-1} \sqrt{x(x+3)} + \cos^{-1} \sqrt{x^2 + 3x + 1}$  is

### KEY - LEVEL-V

#### SINGLE CORRECT ANSWER

- |         |         |         |         |
|---------|---------|---------|---------|
| 1. (D)  | 2. (B)  | 3. (B)  | 4. (C)  |
| 5. (C)  | 6. (D)  | 7. (B)  | 8. (C)  |
| 9. (A)  | 10. (A) | 11. (C) | 12. (A) |
| 13. (A) | 14. (B) | 15. (D) | 16. (B) |
| 17. (D) | 18. (C) | 19. (C) | 20. (A) |
| 21. (A) | 22. (A) | 23. (D) |         |
| 24. (C) | 25. (D) |         |         |

#### MULTIPLE CORRECT ANSWER

- |               |             |
|---------------|-------------|
| 26. (A,B,C,D) | 27. (A,B)   |
| 28. (A,B,C)   | 29. (A,B,C) |
| 30. (A,B,C)   | 31. (B,C)   |
| 32. (A,C)     | 33. (A,B)   |
| 34. (A,B,D)   | 35. (C)     |
| 36. (A,C)     | 37. (A,B,C) |
| 38. (A,B,C,D) | 39. (A,B,C) |
| 40. (A,B,C,D) | 41. (A,B,D) |

#### COMPREHENSION

- |         |         |         |         |
|---------|---------|---------|---------|
| 42. (C) | 43. (A) | 44. (D) | 45. (B) |
| 46. (D) | 47. (C) |         |         |

#### MATRIX MATCH TYPE

48. A- S, B-R, C-P, D-Q  
49. A- R, B-S, C-P, D-Q

#### ASSERTION-REASONING

50. (C) 51. (A) 52. (A) 53. (D)

#### INTEGER TYPE

- 54) 5 55) 2 56) 9 57) 4  
58) 1 59) 3

### HINTS - LEVEL-V

#### SINGLE CORRECT ANSWER

1.  $\cos^{-1} \frac{1}{5\sqrt{2}} = \tan^{-1} \frac{\sqrt{(5\sqrt{2})^2 - 1}}{1} = \tan^{-1} 7$

$$\sin^{-1} \frac{4}{\sqrt{17}} = \tan^{-1} \left( \frac{4}{\sqrt{17-16}} \right) = \tan^{-1} 4$$

So the expression is

$$\tan(\tan^{-1} 7 - \tan^{-1} 4) = \tan \left\{ \tan^{-1} \frac{7-4}{1+7 \times 4} \right\}$$

$$= \tan \tan^{-1} \frac{3}{29} = \frac{3}{29}$$

2.  $\frac{\pi}{2} - \cos^{-1} \cos \left( \frac{2(x^2 + 5|x| + 3) - 2}{x^2 + 5|x| + 3} \right) =$

$$\cot \cot^{-1} \left( \frac{2}{9|x|} - 2 \right) + \frac{\pi}{2}$$

$$\frac{\pi}{2} - 2 + \frac{2}{x^2 + 5|x| + 3} = \frac{2}{9|x|} - 2 + \frac{\pi}{2}$$

$$|x|^2 - 4|x| + 3 = 0 \Rightarrow |x| = 1, 3$$

3. put  $x = \sin \theta$

4.  $\sqrt{1+x^2} \left[ \left( x \cos \cos^{-1} \frac{x}{\sqrt{1+x^2}} + \right. \right.$

$$\left. \sin \cot^{-1} \frac{x}{\sqrt{1+x^2}} \right)^2 - 1 \Bigg]^{1/2}$$

$$\sqrt{1+x^2} \left[ \left( x \cos \cot^{-1} x + \sin \cot^{-1} x \right)^2 - 1 \right]$$

$$= \sqrt{1+x^2} (x^2 + 1 - 1)^{1/2} = x\sqrt{1+x^2}$$

5. As  $0 \leq (\sin^{-1} a)^2 \leq \frac{\pi^2}{4}$ ,  
 $0 \leq (\cos^{-1} b)^2 \leq \pi^2$ ,  $0 \leq (\sec^{-1} c)^2 \leq \pi^2$

$$\left( \text{except } \frac{\pi^2}{4} \right) \text{ and } 0 < (\operatorname{cosec}^{-1} d)^2 \leq \frac{\pi^2}{4}$$

$$\text{So } 0 < (\sin^{-1} a)^2 + (\cos^{-1} b)^2 + (\sec^{-1} c)^2$$

$$+ (\operatorname{cosec}^{-1} d)^2 \leq \frac{5\pi^2}{2}$$

$$\therefore (\sin^{-1} a)^2 + (\cos^{-1} b)^2 + (\sec^{-1} c)^2 +$$

$$(\operatorname{cosec}^{-1} d)^2 = \frac{5\pi^2}{2} \text{ (Given)}$$

$$\Rightarrow (\sin^{-1}a)^2 = \frac{\pi^2}{4}, (\cos^{-1}b)^2 = \pi^2, (\sec^{-1}c)^2 =$$

$$\pi^2 \text{ and } (\operatorname{cosec}^{-1}d)^2 = \frac{\pi^2}{4}$$

$$\text{Hence } (\sin^{-1}a)^2 - (\cos^{-1}b)^2 + (\sec^{-1}c)^2 - (\operatorname{cosec}^{-1}d)^2 = 0$$

$$6. \text{ We have } b \sin^{-1} x + b \cos^{-1} x = \frac{b\pi}{2} \dots\dots (1)$$

$$\text{and } a \sin^{-1} x - b \cos^{-1} x = c \dots\dots (2)$$

(given)

$\therefore$  On adding (1) and (2),

$$\text{we get } (a+b) \sin^{-1} x = \frac{b\pi}{2} + c$$

$$\Rightarrow \sin^{-1} x = \frac{\frac{b\pi}{2} + c}{a+b}$$

$$\text{Similarly } \cos^{-1} x = \frac{\frac{a\pi}{2} - c}{a+b}$$

$$\text{Hence } (a \sin^{-1} x + b \cos^{-1} x) = \frac{\pi ab + c(a-b)}{a+b}$$

$$7. T_n = \cot^{-1} \left( n^2 + \frac{3}{4} \right) = \tan^{-1} \left( \frac{1}{n^2 + \frac{3}{4}} \right)$$

$$= \tan^{-1} \left( n + \frac{1}{2} \right) - \tan^{-1} \left( n - \frac{1}{2} \right)$$

$$S_n = \sum_{n=1}^n t_n = \tan^{-1} \left( n + \frac{1}{2} \right) - \tan^{-1} \frac{1}{2}$$

$$S_\infty = \tan^{-1}(\infty) - \tan^{-1} \frac{1}{2} = \frac{\pi}{2} - \tan^{-1} \frac{1}{2}$$

$$= \cot^{-1} \frac{1}{2} = \tan^{-1} 2$$

$$8. \frac{1}{1 - \sin(\cos^{-1} x)} = 2$$

$$\Rightarrow \frac{1}{2} = 1 - \sin(\cos^{-1} x)$$

$$\Rightarrow \sin(\cos^{-1} x) = \frac{1}{2} \Rightarrow x = \frac{\sqrt{3}}{2}$$

## INVERSE TRIGONOMETRIC FUNCTIONS

$$9. \log_2 x \geq 0 \Rightarrow x \geq 1$$

$$\text{let } x = \tan \theta \Rightarrow \tan \theta \geq 1 \Rightarrow \theta \in \left[ \frac{\pi}{4}, \frac{\pi}{2} \right)$$

$$\lambda = \log_{\frac{1}{\pi}} \left( \sin^{-1}(\sin 2\theta) + 2 \tan^{-1}(\tan \theta) \right)$$

$$= \log_{\frac{1}{\pi}} \left( \sin^{-1}(\sin 2\theta) + 2\theta \right)$$

$$10. 2x = \tan(2 \tan^{-1} a) + 2 \tan(\tan^{-1} a + \tan^{-1} a^3)$$

$$= \frac{2a}{1-a^2} + \frac{2(a+a^3)}{1-a^4} = \frac{2a}{1-a^2} + \frac{2a}{1-a^2} = \frac{4a}{1-a^2}$$

$$\therefore x = \frac{2a}{1-a^2} \Rightarrow a^2 x + 2a = x$$

$$11. \text{ Let } \cos^{-1}(1-x) = \alpha \Rightarrow \cos \alpha = 1-x,$$

$$\cos^{-1} x = \beta \Rightarrow \cos \beta = x$$

$$\therefore \alpha - 2\beta = \frac{\pi}{2} \Rightarrow \alpha - \frac{\pi}{2} = 2\beta$$

introduce sin on both sides and proceed

$$12. \text{ Put } x = \cos \theta \text{ then } \cos \theta = \tan \theta$$

$$\Rightarrow \cos^2 \theta = \sin \theta \Rightarrow \sin^2 \theta + \sin \theta - 1 = 0$$

$$\Rightarrow \sin \theta = \frac{\sqrt{5}-1}{2}$$

$$13. \text{ For } \cos^{-1}(1-x), -1 \leq 1-x \leq 1 \Rightarrow 0 \leq x \leq 2$$

..... (1).

$$\text{for } \cos^{-1} x, -1 \leq x \leq 1 \dots\dots\dots (2).$$

from (1) & (2)  $0 \leq x \leq 1$  for L.H.S=R.H.S

$\cos^{-1}(1-x) = 0$  &  $\cos^{-1} x = 0$  which is not possible

$$14. \frac{\pi}{2} + \sin^{-1} \sqrt{1-x^2} = \cot^{-1} \frac{\sqrt{1-x^2}}{x}$$

$$\tan^{-1} \frac{\sqrt{1-x^2}}{x} + \sin^{-1} \sqrt{1-x^2} = 0$$

$$\Rightarrow x \in [-1, 0) \cup \{1\}$$

$$15. 0 \leq x^2 + x + 1 \leq 1 \text{ and } 0 \leq x^2 + x \leq 1$$

$$\therefore x = -1, 0$$

## INVERSE TRIGONOMETRIC FUNCTIONS

For  $x = -1$

$$\text{L.H.S} = 2\sin^{-1}1 + \cos^{-1}0 = \frac{3\pi}{2}$$

$\therefore x = -1$  is a solution

$$\text{For } x=0, \text{LHS} = 2\sin^{-1}1 + \cos^{-1}0 = \frac{3\pi}{2}$$

Therefore,  $x=0$  is a solution and sum of the solutions  $= -1$

16. Given that  $\sin^{-1}x + \tan^{-1}x = 2k + 1$

The range of the function  $\sin^{-1}x + \tan^{-1}x$  is

$$\left[ -\frac{3\pi}{4}, \frac{3\pi}{4} \right],$$

[as both functions are increasing]

Therefore, the integral values of  $k$  are  $-1$  and  $0$

17.  $\sin[\cot^{-1}(1+x)] = \cos(\tan^{-1}x)$

$$\Rightarrow \sin\left\{\sin^{-1}\frac{1}{\sqrt{2+2x+x^2}}\right\} = \cos\left\{\cos^{-1}\frac{1}{\sqrt{1+x^2}}\right\}$$

$$\Rightarrow \frac{1}{\sqrt{2+2x+x^2}} = \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow x = -\frac{1}{2}$$

18. From the given equation

$$\sin^2\theta - 2\sin\theta + 3 = 5^{\sec^2\theta} + 1 \text{ we get}$$

$$(\sin\theta - 1)^2 + 2 = 5^{\sec^2\theta} + 1$$

$$\text{LHS} \leq 6, \text{RHS} \geq 6$$

Possible solution  $\sin\theta = -1$  when  $\text{LHS} = \text{RHS}$

$$\Rightarrow \cos^2\theta = 0 \Rightarrow \cos^2\theta - \sin\theta = 1$$

19.  $\cos^{-1}x < \sin 1 \Rightarrow x > \cos(\sin 1)$

$$\cos^{-1}x > \cos 1 \Rightarrow x < \cos(\cos 1)$$

20.  $x^2 - \lambda x + \sin^{-1}\sin(\pi - 4) > 0 \quad \forall x \in R$

$$\Rightarrow x^2 - \lambda x + \pi - 4 > 0 \quad \forall x \in R$$

$$\Rightarrow \lambda^2 - 4(\pi - 4) < 0$$

$$\therefore \lambda \in \phi$$

21.  $(\sec^{-1}x - 4)(\sec^{-1}x - 2) > 0$

$$\therefore \sec^{-1}x < 2 \text{ and } \sec^{-1}x > 4$$

## JEE ADVANCED - VOL - II

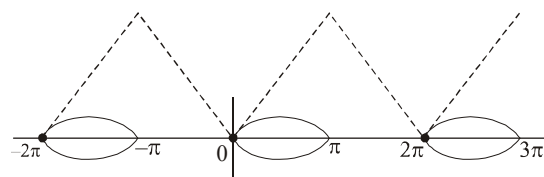
22. As domain of first function is  $\left[-\frac{1}{2}, 0\right) \cup \left\{\frac{1}{2}\right\}$

and domain of 2<sup>nd</sup> function  $[-3, -2]$  there is no common point

23. [Given expression]

$$\begin{aligned} &= \left\{ \frac{\pi}{2} - \cos^{-1}(\cos 2) \right\} - \left\{ \frac{\pi}{2} - \sin^{-1}(\sin 2) \right\} \\ &+ \left\{ \frac{\pi}{2} - \cot^{-1}(\cot 4) \right\} - \left\{ \frac{\pi}{2} - \tan^{-1}(\tan 4) \right\} \\ &+ \left\{ \frac{\pi}{2} - \operatorname{cosec}^{-1}(\operatorname{cosec} 6) \right\} - \left\{ \frac{\pi}{2} - \sec^{-1}(\sec 6) \right\} \\ &= \sin^{-1}(\sin 2) - \cos^{-1}(\cos 2) + \tan^{-1}(\tan 4) \\ &- \cot^{-1}(\cot 4) + \sec^{-1}(\sec 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 6) \\ &= (\pi - 2) - 2 + (4 - \pi) - (4 - \pi) + (2\pi - 6) \\ &- (6 - 2\pi) = \pi - 4 + 4\pi - 12 = 5\pi - 16. \end{aligned}$$

24. Graph of  $y = \cos^{-1}(\cos x)$  &  $|y| = \sin x$



$$(x, y) = (0, 0), (2\pi, 0), (-2\pi, 0)$$

$\therefore$  Number of solutions is 3.

25. For domain of  $f(x)$ , we must have  $-1 \leq [x] \leq 1$

$$\Rightarrow -1 \leq x < 2, \text{ so set } A = [-1, 2)$$

$$f(x) = \frac{2}{\pi} \left( \sin^{-1}[x] + \frac{\pi}{2} \right)$$

$$\left( \text{As } \tan^{-1}[x] + \cot^{-1}[x] = \frac{\pi}{2} \quad \forall x \in A \right)$$

So, set  $B = \{0, 1, 2\} = \text{Range of } f(x)$

$$\text{Now } A \cup B = [-1, 2) \cup \{0, 1, 2\} = [-1, 2]$$

Hence number of integers in  $(A \cup B) = 4$

## MULTIPLE CORRECT ANSWER

26. As  $-\frac{\pi}{2} \leq \sin^{-1}x \leq \frac{\pi}{2} \quad \forall -1 \leq x \leq 1$

$$0 \leq (\sin^{-1}x) \leq \frac{\pi^2}{4}$$

$$\therefore 0 \leq (\sin^{-1} x)^2 + (\sin^{-1} y)^2 + (\sin^{-1} z)^2 \leq \frac{3\pi^2}{4}$$

$$\therefore (\sin^{-1} x)^2 + (\sin^{-1} y)^2 + (\sin^{-1} z)^2 = \frac{3\pi^2}{4} \text{ is}$$

possible if  $x, y, z \in \{-1, 1\}$

$\therefore$  Possible values of  $x - y + z$  from the ordered triplet  $(x, y, z)$  are as follows :

| $(x, y, z)$    | $x - y + z$ |
|----------------|-------------|
| $(-1, -1, -1)$ | $-1$        |
| $(-1, 1, 1)$   | $-1$        |
| $(1, -1, 1)$   | $3$         |
| $(1, 1, -1)$   | $-1$        |
| $(1, 1, 1)$    | $1$         |
| $(1, -1, -1)$  | $1$         |
| $(-1, 1, -1)$  | $-3$        |
| $(-1, -1, 1)$  | $1$         |

Hence set of values of  $x - y + z$  is  $\{\pm 1, \pm 3\}$

$$-1 \leq x \leq 1$$

$$27. (A) \frac{-\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4} \Rightarrow \frac{-\pi}{2} \leq 2 \tan^{-1} x \leq \frac{\pi}{2}$$

$$\frac{-1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}} \Rightarrow \frac{-\pi}{4} < \sin^{-1} x < \frac{\pi}{4}$$

$$(B) \Rightarrow \frac{-\pi}{2} < 2 \sin^{-1} x < \frac{\pi}{2}$$

$$28. 2 \tan^{-1} \frac{1}{3} = \tan^{-1} \frac{3}{4} = \sin^{-1} \frac{3}{5} = \frac{\pi}{2} - \cos^{-1} \frac{3}{5} = \frac{\pi}{2} - \sin^{-1} \frac{4}{5}$$

$$\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{2} = \frac{\pi}{4} \Rightarrow \tan^{-1} \frac{1}{3} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2} = \frac{\pi}{4} - \cot^{-1} 2$$

$$29.(A) \sin(\tan^{-1} 3 + \cot^{-1} 3) = \sin \pi / 2 = 1$$

$$30. \theta = \tan^{-1}(2 \tan^2 \theta) - \tan^{-1}\left(\frac{1}{3} \tan \theta\right)$$

$$\Rightarrow \tan \theta = \frac{2 \tan^2 \theta - \frac{1}{3} \tan \theta}{1 + \frac{2}{3} \tan^3 \theta}$$

$$1 = \left( \frac{2 \tan \theta - \frac{1}{3}}{1 + \frac{2}{3} \tan^3 \theta} \right) \Rightarrow \tan \theta = 1, -2$$

## INVERSE TRIGONOMETRIC FUNCTIONS

$$31. f\left(\frac{8\pi}{9}\right) = e^{\cos^{-1}\left(\sin\left(\frac{11\pi}{9}\right)\right)} = e^{\cos^{-1}\cos\frac{13\pi}{18}} = e^{\frac{13\pi}{18}}$$

$$f\left(\frac{-7\pi}{4}\right) = e^{\cos^{-1}\sin\left(\frac{-7\pi}{4} + \frac{\pi}{3}\right)} = e^{\cos^{-1}\cos\frac{\pi}{12}} = e^{\frac{\pi}{12}}$$

$$32. \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad \forall x \in [-1, 1]$$

$$33. \sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$$

$$\Rightarrow \frac{\pi}{2} - \cos^{-1} x + \frac{\pi}{2} - \cos^{-1}(1-x) = \cos^{-1} x$$

$$\Rightarrow 2\cos^{-1} x = \pi - \cos^{-1}(1-x)$$

$$\Rightarrow \cos^{-1}(2x^2 - 1) = \cos^{-1}(x - 1)$$

$$\Rightarrow 2x^2 - x = 0$$

$$\Rightarrow x(2x - 1) = 0$$

$$\therefore x = 0, \frac{1}{2}$$

$$34. \tan^{-1}(x-1) + \tan^{-1}(x)$$

$$+ \tan^{-1}(x+1) = \tan^{-1} 3x$$

$$\Rightarrow \tan^{-1}(x-1) + \tan^{-1}(x)$$

$$= \tan^{-1} 3x - \tan^{-1}(x+1)$$

$$\Rightarrow \tan^{-1} \left[ \frac{(x-1)+x}{1-(x-1)(x)} \right] = \tan^{-1} \left[ \frac{3x-(x+1)}{1+3x(x+1)} \right]$$

$$\Rightarrow \frac{2x-1}{1-x^2+x} = \frac{2x-1}{1+3x^2+3x}$$

$$\Rightarrow (1-x^2+x)(2x-1) = (1+3x^2+3x)(2x-1)$$

$$\Rightarrow x = 0, \pm \frac{1}{2}$$

$$35. \sin^{-1}(\sin x) = x; \frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$= \pi - x; \frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$$

$$= -2\pi + x; \frac{3\pi}{2} \leq x \leq \frac{5\pi}{2}$$

$$= 3\pi - x; \frac{5\pi}{2} \leq x \leq \frac{7\pi}{2}$$

$$\cos^{-1}(\cos x) = x; 0 \leq x < \pi$$

$$= 2\pi - x; \pi \leq x \leq 2\pi$$

$$36. \tan^{-1} \left\{ \frac{\left( \frac{x-1}{x-2} \right) + \left( \frac{x+1}{x+2} \right)}{1 - \left( \frac{x-1}{x-2} \right) \left( \frac{x+1}{x+2} \right)} \right\} = \frac{\pi}{4}$$

$$\tan^{-1} \left\{ \frac{\frac{2(x^2-2)}{(x^2-4)}}{1 - \left( \frac{x^2-1}{x^2-4} \right)} \right\} = \frac{\pi}{4}$$

$$\text{Also } \left( \frac{x^2-1}{x^2-4} \right) > 1$$

Hence, we take the LHS

$$\pi + \tan^{-1} \left\{ \frac{2x^2-4}{-3} \right\} = \frac{\pi}{4}$$

$$\Rightarrow \pi - \frac{\pi}{4} = \tan^{-1} \left\{ \frac{2x^2-4}{-3} \right\}$$

$$\therefore \tan \left( \frac{3\pi}{4} \right) = \frac{2x^2-4}{-3} \Rightarrow -1 = \frac{2x^2-4}{-3}$$

$$\Rightarrow 2x^2 = 1$$

$$\therefore x = \pm \frac{1}{\sqrt{2}}$$

$$37. y = \sqrt{y} \Rightarrow y = 0 \text{ or } y = 1$$

$$\sin^{-1} |\sin x| = 1 \Rightarrow x = 1 \text{ or } \pi - 1; \pi + 1$$

$$38. \text{ The domain holds only for } x = -1$$

Now,

$$\pi/2 + \tan^{-1}(x^2 - 3x - k^2) > \pi/2 \text{ for } x = -1$$

$$\Rightarrow k^2 - 4 < 0.$$

$$39. \text{ (A) } \cos(\tan^{-1}(\tan(4 - \pi))) \\ = \cos(4 - \pi) = \cos(\pi - 4) = -\cos 4 > 0$$

$$\text{ (B) } \sin(\cot^{-1}(\cot(4 - \pi))) \\ = \sin(4 - \pi) = -\sin 4 > 0 \text{ (as } \sin 4 < 0)$$

$$\text{ (C) } \tan(\cos^{-1}(\cos(2\pi - 5))) \\ = \tan(2\pi - 5) = -\tan 5 > 0 \text{ (as } \tan 5 < 0)$$

$$\text{ (D) } \cot(\sin^{-1}(\sin(\pi - 4))) \\ = \cot(\pi - 4) = -\cot 4 < 0$$

$$40. |\tan^{-1} x| = \begin{cases} \tan^{-1} x & \text{if } 0 \leq \tan^{-1} x < \frac{\pi}{2} \\ -\tan^{-1} x & \text{if } -\frac{\pi}{2} < \tan^{-1} x < 0 \end{cases}$$

$$= \begin{cases} \tan^{-1} x, & x \geq 0 \\ -\tan^{-1} x, & x < 0 \end{cases}$$

$$|\tan^{-1} x| = \tan^{-1} |x|, \forall x \in R$$

$$\tan |\tan^{-1} x| = \tan \tan^{-1} |x| = |x|$$

$$\text{if } x = -1 \text{ then } \cot |\cot^{-1} x| = 1$$

$$41. \text{ (A) } \left[ \min_{x \in R} e^x \right] = 0, \left[ e^{\frac{-\pi}{2}} \right] = 0$$

$$[\sin(\sin x)] = 0, x \in (0, \frac{\pi}{2})$$

$$\text{ (B) } [\cos(\cos x)] = 0, x \in (0, \frac{\pi}{2})$$

$$\text{ (C) } 1 > \frac{\pi}{2} - \sin 1 \\ \sin 1 < \cos(\sin 1)$$

$$\frac{\pi}{2} < 2 \sin^{-1} \frac{1}{4} \Rightarrow \frac{\pi}{4} < \sin^{-1} \frac{1}{4}$$

$$\text{ (D) } \frac{1}{\sqrt{2}} < \frac{1}{4}$$

## COMPREHENSION

42-44. For

$$x \in \left[ -1, -\frac{1}{2} \right], \cos^{-1}(4x^3 - 3x) = 3\cos^{-1} x - 2\pi$$

$$\Rightarrow a = -2\pi \text{ and } b = 3 \Rightarrow a + b\pi = \pi$$

For

$$x \in \left[ -\frac{1}{2}, \frac{1}{2} \right], \cos^{-1}(4x^3 - 3x) = 2\pi - 3\cos^{-1} x$$

$$\Rightarrow a = 2\pi \text{ and}$$

$$b = -3 \Rightarrow \sin^{-1} \left( \sin \frac{a}{b} \right) = \sin^{-1} \left( \sin \frac{2\pi}{-3} \right)$$

$$= \sin^{-1} \left( -\frac{\sqrt{3}}{2} \right) = -\frac{\pi}{3}$$

$$\text{For } x \in \left( \frac{1}{2}, 1 \right], \quad \cos^{-1}(4x^3 - 3x) = 3 \cos^{-1} x$$

$$\Rightarrow a = 0 \text{ and } b = 3$$

$$\therefore \lim_{y \rightarrow a} b \cos(y) = \lim_{y \rightarrow 0} 3 \cos(y) = 3.$$

$$45-47. \text{ Let } \cos^{-1} x = a \Rightarrow a \in [0, \pi]$$

$$\text{and } \sin^{-1} y = b \Rightarrow b \in [-\pi/2, \pi/2]$$

$$\text{We have } a + b^2 = \frac{p\pi^2}{4} \quad (\text{i})$$

$$\text{and } ab^2 = \frac{\pi^4}{16} \quad (\text{ii})$$

Since

$$b^2 \in [0, \pi^2/4], \text{ we get } a + b^2 \in [0, \pi + \pi^2/4]$$

$$\text{So, from Eq(i) we get } 0 \leq \frac{p\pi^2}{4} \leq \pi + \frac{\pi^2}{4}$$

$$\text{i.e., } 0 \leq p \leq \frac{4}{\pi} + 1$$

Since  $p \in \mathbb{Z}$ , so  $p = 0, 1$  or  $2$

Substituting the value of  $b^2$  from Eq (i) in Eq(ii) we get

$$a \left( \frac{p\pi^2}{4} - a \right) = \frac{\pi^4}{16} \Rightarrow 16a^2 - 4p\pi^2 a + \pi^4 = 0 \quad (\text{iii})$$

Since  $a \in \mathbb{R} \Rightarrow D \geq 0$

$$\text{i.e., } 16p^2 \pi^4 - 64\pi^4 \geq 0$$

$$\Rightarrow p^2 \geq 4 \Rightarrow p \geq 2 \Rightarrow p = 2$$

Substituting  $p=2$  in Eq (iii), we get

$$16a^2 - 8\pi^2 a + \pi^4 = 0$$

$$\Rightarrow (4a - \pi^2)^2 = 0$$

$$\Rightarrow a = \frac{\pi^2}{4} = \cos^{-1} x \Rightarrow x = \cos \frac{\pi^2}{4}$$

From Eq. (ii) we get

$$\frac{\pi^2}{4} b^2 = \frac{\pi^4}{16} \Rightarrow b = \pm \frac{\pi}{2} = \sin^{-1} y \Rightarrow y = \pm 1$$

### MATRIX MATCH TYPE

$$48. \quad (\text{i}) \lambda = \mu = \nu = -1$$

### INVERSE TRIGONOMETRIC FUNCTIONS

$$(\text{ii}) x_i = 1 \text{ for } i = 1, 2, \dots, 10$$

$$(\text{iii}) \sin^{-1}(x_1) + \sin^{-1}(x_2) + \dots + \sin^{-1}(x_{2n})$$

$$= 2n \cdot \frac{\pi}{2}, \Rightarrow x_i = 1$$

$$(\text{iv}) \text{ put } x = \sin q \text{ \& } \frac{1}{2} = \sin a$$

$$49. \quad \text{A) } \sin \left[ \frac{\pi}{3} + \frac{\pi}{6} \right] = \sin \frac{\pi}{2} = 1$$

$$\text{B) } \cos \left[ \frac{5\pi}{6} + \frac{\pi}{6} \right] = \cos \pi = -1$$

$$\text{C) let } \theta = \cos^{-1} \frac{\sqrt{5}}{3} \text{ . find } \tan \frac{\theta}{2}$$

$$\text{D) let } \theta = \sin^{-1} \frac{4}{5} \text{ . find } \tan 2\theta$$

### ASSERTION-REASONING TYPE

$$50. \quad \cos ec^{-1} x > \sec^{-1} x$$

$$\Rightarrow \cos ec^{-1} x > \frac{\pi}{2} - \cos ec^{-1} x$$

$$\Rightarrow \cos ec^{-1} x > \frac{\pi}{4}$$

$$\Rightarrow 1 \leq x < \sqrt{2} \text{ and } \left( \frac{1}{2} + \frac{1}{\sqrt{2}} \right) \in [1, \sqrt{2})$$

$$51. \quad \tan^{-1} \left( \frac{2+3}{1-6} \right) = \frac{3\pi}{4}$$

$$52. \quad \sin^{-1} x = \tan^{-1} \frac{x}{\sqrt{1-x^2}} > \tan^{-1} x > \tan^{-1} y$$

$$\left[ \because x > y, \frac{x}{\sqrt{1-x^2}} > x \right]$$

There fore, statement 2 is true

$$\text{Now, } e < \pi \Rightarrow \frac{1}{\sqrt{e}} > \frac{1}{\sqrt{\pi}}$$

By statement 2, we have

$$\sin^{-1} \left( \frac{1}{\sqrt{e}} \right) > \tan^{-1} \left( \frac{1}{\sqrt{e}} \right) > \tan^{-1} \left( \frac{1}{\sqrt{\pi}} \right)$$

There fore, statement 1 is true

$$53. \quad \tan^{-1} \frac{1}{x} = \cot^{-1} x - \pi, \quad (x < 0)$$

INTEGER TYPE

$$54. \quad (\cot^{-1} x)(\tan^{-1} x) + \left(2 - \frac{\pi}{2}\right) \cot^{-1} x$$

$$-3 \tan^{-1} x - 3 \left(2 - \frac{\pi}{2}\right) > 0$$

$$\Rightarrow (\cot^{-1} x - 3) \left(2 - \cot^{-1} x\right) > 0$$

$$\Rightarrow (\cot^{-1} x - 3) (\cot^{-1} x - 2) < 0$$

$$\Rightarrow 2 < \cot^{-1} x < 3$$

$$\Rightarrow \cot 3 < x < \cot 2$$

(as  $\cot^{-1} x$  is a decreasing function)

$$\Rightarrow \text{Hence, } x \in (\cot 3, \cot 2)$$

$$\Rightarrow \cot^{-1} a + \cot^{-1} b = \cot^{-1} (\cot 3) + \cot^{-1} (\cot 2) = 5$$

$$55. \quad \text{Since } \sin^{-1} \text{ is defined for } [-1, 1]$$

$$\therefore a = 0$$

$$\therefore x = \sin^{-1} 1 + \cos^{-1} 1 - \tan^{-1} 1 = \frac{\pi}{4}$$

$$\Rightarrow \sec^2 x = 2$$

$$56. \quad 1 + \sin(\cos^{-1} x) + \sin^2(\cos^{-1} x) + \dots \infty = 2$$

$$\Rightarrow \frac{1}{1 - \sin(\cos^{-1} x)} = 2$$

$$\Rightarrow \frac{1}{2} = 1 - \sin(\cos^{-1} x)$$

$$\Rightarrow \sin(\cos^{-1} x) = \frac{1}{2} \Rightarrow \cos^{-1} x = \frac{\pi}{6}$$

$$\Rightarrow x = \frac{\sqrt{3}}{2} \Rightarrow 12x^2 = 9$$

$$57. \quad f(x) = \sin^{-1} x + 2 \tan^{-1} x + (x+2)^2 - 3$$

Domain of  $f(x)$  is  $[-1, 1]$

Also  $f(x)$  is an increasing function in the domain

$$\therefore p = f_{\min}(x) = f(-1)$$

$$= -\frac{\pi}{2} + 2 \left( \frac{-\pi}{4} \right) + 1 - 3 = -\pi - 2$$

and

$$q = f_{\max}(x) = f(1) = \frac{\pi}{2} + 2 \left( \frac{\pi}{4} \right) + 9 - 3 = \pi + 6$$

There fore, the range of  $f(x)$  is  $[-\pi - 2, \pi + 6]$

$$58. \quad \text{Let us assume } \tan^{-1} \sqrt{\frac{xr}{yz}} = \alpha, \quad \tan^{-1} \sqrt{\frac{yr}{zx}} = \beta \text{ and}$$

$$\tan^{-1} \sqrt{\frac{zr}{xy}} = \gamma$$

Then we have L.H.S.

$$= \tan^{-1} \alpha + \tan^{-1} \beta + \tan^{-1} \gamma = \tan^{-1} \left[ \frac{\alpha + \beta + \gamma - \alpha\beta\gamma}{1 - \alpha\beta - \beta\gamma - \gamma\alpha} \right]$$

Now evaluating

$$\alpha + \beta + \gamma - \alpha\beta\gamma = \sqrt{\frac{xr}{zy}} + \sqrt{\frac{yr}{zx}} + \sqrt{\frac{zr}{xy}} - \sqrt{\frac{xr}{zy}} \sqrt{\frac{yr}{zx}} \sqrt{\frac{zr}{xy}}$$

$$= \frac{x\sqrt{r} + y\sqrt{r} + z\sqrt{r}}{\sqrt{xyz}} - \frac{r\sqrt{r}}{\sqrt{xyz}}$$

$$= \frac{r\sqrt{r}}{\sqrt{xyz}} - \frac{r\sqrt{r}}{\sqrt{xyz}} = 0 \quad [\because x + y + z = r]$$

Also,  $1 - \alpha\beta - \beta\gamma - \gamma\alpha$

$$= 1 - \sqrt{\frac{xr}{yz}} \sqrt{\frac{yr}{zx}} - \sqrt{\frac{yr}{zx}} \sqrt{\frac{zr}{xy}} - \sqrt{\frac{zr}{xy}} \sqrt{\frac{xr}{yz}}$$

$$= 1 - \frac{r}{z} - \frac{r}{x} - \frac{r}{y} = 1 - r \left[ \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right] \neq 0$$

$\therefore$  LHS=RHS only for  $k = 1$

$$59. \quad \text{We must have } x(x+3) \geq 0$$

$$\Rightarrow x \geq 0 \text{ or } x \leq -3$$

$$\text{Also, } -1 \leq x^2 + 3x + 1 \leq 1$$

$$\Rightarrow x(x+3) \leq 0 \Rightarrow -3 \leq x \leq 0$$

From Eqs. (i) and (ii), we get  $x = \{0, -3\}$

Hence, required sum is 3

**LEVEL-VI****SINGLE ANSWER QUESTIONS**

1. The range of value of  $p$  for which the equation  $\sin \cos^{-1}(\cos(\tan^{-1} x)) = p$  has a solution is

A)  $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$  B)  $[0, 1)$   
 C)  $\left(\frac{1}{\sqrt{2}}, 1\right)$  D)  $(-1, 1)$

2. The solution set of the equation

$$\sin^{-1} \sqrt{1-x^2} + \cos^{-1} x = \cot^{-1} \frac{\sqrt{1-x^2}}{x} - \sin^{-1} x$$

is

A)  $[-1, 1] - \{0\}$  B)  $(0, 1] \cup \{-1\}$   
 C)  $[-1, 0) \cup \{1\}$  D)  $[-1, 1]$

3. The solution set of the equation

$$2 \cos^{-1} x = \cot^{-1} \left( \frac{2x^2 - 1}{2x\sqrt{1-x^2}} \right) \text{ is}$$

A)  $(0, 1)$  B)  $(-1, 1) - \{0\}$   
 C)  $(-1, 0)$  D)  $[-1, 1]$

4. Exhaustive set of parameter 'a' so that  $\sin^{-1} x - \tan^{-1} x = a$  has a solution, is

(A)  $\left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$  (B)  $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$   
 (C)  $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$  (D)  $\left[-\frac{\pi}{12}, \frac{\pi}{12}\right]$

5. The sum of roots of the equation

$$\cos^{-1}(\cos x) = [x] \text{ (where } [.] \text{ is G.I.F.) is}$$

A)  $2\pi + 3$  B)  $\pi + 3$  C)  $\pi - 3$  D)  $2\pi - 3$

6. The product of all values of  $x$  satisfying the

$$\text{equation } \sin^{-1} \cos \left( \frac{2x^2 + 10|x| + 4}{x^2 + 5|x| + 3} \right)$$

**INVERSE TRIGONOMETRIC FUNCTIONS**

$$= \cot \left( \cot^{-1} \left( \frac{2-18|x|}{9|x|} \right) \right) + \frac{\pi}{2} \text{ is}$$

A) 9 B) -9 C) -3 D) -1

7. If  $\left[ \sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x \right] = 1$ , where  $[.]$  denotes the greatest integer function, then  $x$  is given by the interval

(A)  $[\tan \sin \cos 1, \tan \sin \cos \sin 1]$   
 (B)  $(\tan \sin \cos 1, \tan \sin \cos \sin 1)$   
 (C)  $[-1, 1]$   
 (D)  $[\sin \cos \tan 1, \sin \cos \sin \tan 1]$

8. If  $\sin^{-1} \left( \sin \left( \frac{2x^2 + 4}{1+x^2} \right) \right) < \pi - 3$  then  $x$  satisfies

(A)  $|x| > 1$  (B)  $|x| < 1$  (C)  $|x| \leq 1$  (D)  $|x| \geq 1$

9. If  $2^{\frac{2\pi}{\sin^{-1} x}} - 2(a+2)2^{\frac{\pi}{\sin^{-1} x}} + 8a < 0$  for at least one real  $x$  then

A)  $\frac{1}{8} \leq a < 2$  B)  $a < 2$   
 C)  $a \in \mathbb{R} - \{2\}$  D)  $a \in \left[0, \frac{1}{8}\right) \cup (2, \infty)$

10. The complete solution set of  $[\tan^{-1} x]^2 - 8[\tan^{-1} x] + 16 \leq 0$ , where  $[.]$  denotes the greatest integer function, is

(A)  $(-\infty, \tan 4]$  (B)  $[\tan 4, \tan 3]$   
 (C)  $(\tan 5, \infty)$  (D) Does not exist

11. If  $\cot^{-1} \left( \frac{n^2 - 10n + 26}{2\sqrt{3}} \right) > \frac{\pi}{6}$ ,  $n \in \mathbb{N}$ , then the minimum value of  $n$  is

A) 3 B) 5 C) 6 D) 7

**MULTIANSWER QUESTIONS**

12. Identify the pair(s) of functions which are identical

(A)  $y = \tan(\cos^{-1} x)$ ;  $y = \frac{\sqrt{1-x^2}}{x}$



## INVERSE TRIGONOMETRIC FUNCTIONS

(B)  $y = \tan(\cot^{-1} x); y = \frac{1}{x}$

(C)  $y = \sin(\arctan x); y = \frac{x}{\sqrt{1+x^2}}$

(D)  $y = \cos(\tan^{-1} x); y = \sin(\cot^{-1} x)$

13. If  $y$  is infinite and  $\tan^{-1} y = 4 \tan^{-1} x$  then

(A)  $x^2 = 3 + 2\sqrt{2}$  (B)  $x^2 = 3 - 2\sqrt{2}$

(C)  $x^4 = 6x^2 - 1$  (D)  $x^4 = 6x^2 + 1$

14. If the equation  $\sin^{-1}(x^2+x+1) + \cos^{-1}(\lambda x+1) = \pi/2$  has exactly two solutions, then  $\lambda$  cannot have the integral value

(A) -1 (B) 0 (C) 1 (D) 2

15. The solution of the equation  $\sin [2\cos^{-1} \{\cot (2 \tan^{-1} x)\}] = 0$  are

(A)  $\pm 1$  (B)  $1 \pm \sqrt{2}$  (C)  $-1 \pm \sqrt{2}$  (D)  $\pm \sqrt{2}$

16. If  $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$  and  $\sin 2x = \cos 2y$ , then

(A)  $x = \frac{\pi}{8} + \sqrt{\frac{1}{2} - \frac{\pi^2}{64}}$

(B)  $y = \sqrt{\frac{1}{2} - \frac{\pi^2}{64}} - \frac{\pi}{12}$

(C)  $x = \frac{\pi}{12} + \sqrt{\frac{1}{2} - \frac{\pi^2}{64}}$

(D)  $y = \sqrt{\frac{1}{2} - \frac{\pi^2}{64}} - \frac{\pi}{8}$

17. The value of  $k$  ( $k > 0$ ) such that the length of the longest interval in which the function

$f(x) = \sin^{-1} |\sin kx| + \cos^{-1} (\cos kx)$

is constant  $\frac{\pi}{4}$  is/are

A) 8 B) 4 C) 12 D) 16

## COMPREHENSION TYPE QUESTIONS

### Passage-1:

While defining inverse trigonometric functions, a new system is followed where

## JEE ADVANCED - VOL - II

domains and ranges have been redefined as follows.

| Function      | Domain       | Range                                        |
|---------------|--------------|----------------------------------------------|
| $\sin^{-1} x$ | $[-1, 1]$    | $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ |
| $\tan^{-1} x$ | $\mathbb{R}$ | $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ |
| $\cos^{-1} x$ | $[-1, 1]$    | $[\pi, 2\pi]$                                |
| $\cot^{-1} x$ | $\mathbb{R}$ | $[\pi, 2\pi]$                                |

18.  $\sin^{-1}(-x)$  is equal to

(A)  $-\sin^{-1} x$  (B)  $\pi + \sin^{-1} x$

(C)  $2\pi - \sin^{-1} x$  (D)  $3\pi - \cos^{-1} \sqrt{1-x^2}, x > 0$

19. If  $f(x) = 3\sin^{-1} x - 2\cos^{-1} x$  then  $f(x)$  is

(A) even function (B) odd function

(C) Neither even nor odd

(D) even as well as odd function

20. The minimum of  $(\sin^{-1} x)^3 - (\cos^{-1} x)^3$  is equal to

(A)  $-\frac{63}{8}\pi^3$  (B)  $\frac{63}{8}\pi^3$

(C)  $\frac{125\pi^3}{32}$  (D)  $\frac{125^3}{32}$

21. If in another such system range of  $\sin^{-1} x$  is

$\left[-\frac{5\pi}{2}, -\frac{3\pi}{2}\right]$  and  $\sin^{-1} x + \cos^{-1} x = 5\pi/2$

then the range of  $\cos^{-1} x$  will be

(A)  $[4\pi, 5\pi]$  (B)  $[3\pi, 4\pi]$

(C)  $[6\pi, 7\pi]$  (D)  $[5\pi, 6\pi]$

### Passage-2:

For  $x, y, z, t \in \mathbb{R}$ ,

$\sin^{-1} x + \cos^{-1} y + \sec^{-1} z \geq t^2 - \sqrt{2}\pi t + 3\pi$

22. The value of  $x+y+z$  is equal to

A) 1 B) 0 C) 2 D) -1

23. The principal value of  $\cos^{-1}(\cos 5t^2)$  is

A)  $\frac{3\pi}{2}$  B)  $\frac{\pi}{2}$  C)  $\frac{\pi}{3}$  D)  $\frac{2\pi}{3}$

24. The value of  $\cos^{-1}(\min\{x, y, z\})$  is

A) 0 B)  $\frac{\pi}{2}$  C)  $\pi$  D)  $\frac{\pi}{3}$

## MATRIX MATCH TYPE

25. Let  $t_1 = (\sin^{-1} x)^{\sin^{-1} x}$ ,  $t_2 = (\sin^{-1} x)^{\cos^{-1} x}$ ,

$$t_3 = (\cos^{-1} x)^{\sin^{-1} x}, t_4 = (\cos^{-1} x)^{\cos^{-1} x}$$

Match the items of Column I with the items Column II

Column - I

Column - II

(A)  $x \in (0, \cos 1)$

(p)  $t_1 > t_2 > t_4 > t_3$

(B)  $x \in \left( \cos 1, \frac{1}{\sqrt{2}} \right)$

(q)  $t_4 > t_3 > t_1 > t_2$

(C)  $x \in \left( \frac{1}{\sqrt{2}}, \sin 1 \right)$

(r)  $t_1 > t_2 > t_4 > t_3$

(D)  $x \in (\sin 1, 1)$

(s)  $t_3 > t_4 > t_1 > t_2$

26. Column-I

Column-II

A)  $(\sin^{-1} x)^2 + (\sin^{-1} y)^2 = \frac{\pi^2}{2}$

p) 1

$\Rightarrow x^3 + y^3 =$

B)  $(\cos^{-1} x)^2 + (\cos^{-1} y)^2 = 2\pi^2$

q) -2

$\Rightarrow x^5 + y^5$

C)  $(\sin^{-1} x)^2 (\cos^{-1} y)^2 = \frac{\pi^4}{4} \Rightarrow |x - y|$  r) 0

D)  $|\sin^{-1} x - \sin^{-1} y| = \pi \Rightarrow x^y$  s) 2

27. Let  $(x, y)$  be such that

$$\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}.$$

Match the statements in Column I with statements in column II. (IIT 2007)

Column I

(A) If  $a=1$  and  $b=0$ , then  $(x, y)$

(B) If  $a=1$  and  $b=1$ , then  $(x, y)$

(C) If  $a=1$  and  $b=2$ , then  $(x, y)$

(D) If  $a=3$  and  $b=2$ , then  $(x, y)$

Column II

(p) lies on the circles  $x^2 + y^2 = 1$

## INVERSE TRIGONOMETRIC FUNCTIONS

(q) lies on  $(x^2 - 1)(y^2 - 1) = 0$

(r) lies on  $y = x$

(s) lies on  $(4x^2 - 1)(y^2 - 1) = 0$

## STATEMENT TYPE QUESTIONS

The following questions consist of two statements, one labelled as 'Statement I' and the other 'Statement II'. You have to examine these two statements carefully and decide if the Statement I and the Statement II, are individually true and if so, whether the Statement II is the correct explanation for the given Statement I. Select your answer to these items using the codes given below and then select the correct option.

(A) Both I and II are individually true and II is the correct explanation of I

(B) Both I and II are individually true but II is not the correct explanation of I

(C) I is true but II is false

(D) I is false but II is true

28. Statement-I: If  $x < 0$  then

$$2 \sec^{-1} \sqrt{1+x^2} - \cos^{-1} \frac{1-x^2}{1+x^2} = 0$$

Statement II : If  $x < 0$  then

$$\sec^{-1} \sqrt{1+x^2} = -\tan^{-1} x,$$

$$\cos^{-1} \frac{1-x^2}{1+x^2} = -2 \tan^{-1} x$$

29. Statement 1 :  $\sin^{-1}(\sin 5) > x^2 - 4x$  has many solutions lying in the interval

$$(2 - \sqrt{9-2\pi}, 2 + \sqrt{9-2\pi})$$

Statement 2 :  $\sin^{-1}(\sin 5) = 5 - 2\pi$

30. Statement -1 : Let  $|x| \leq 1$ , then the value of

$$\sin^{-1}[\cos(\sin^{-1} x)] + \cos^{-1}[\sin(\cos^{-1} x)] \text{ is } \pi/2$$

Statement -2 : For the values of  $x \in [-1, 1]$

$$\cos(\sin^{-1} x) = \sin \cos^{-1} x$$

## INVERSE TRIGONOMETRIC FUNCTIONS

31. **Statement I :**  $f(x) = (\cos^{-1} x)^2 + \frac{\pi}{2} \sin^{-1} x$

then Range of  $f(x)$  is  $\left[\frac{3\pi^2}{16}, \frac{3}{4}\pi^2\right]$

**Statement II :** If

$f(x) = ax^2 + bx + c$  and if  $x_1 < \frac{-b}{2a} < x_2$  then

Range of  $f(x)$  in the interval  $[x_1, x_2]$  is

$$\left[ \min \left\{ f(x_1), f(x_2), f\left(-\frac{b}{2a}\right) \right\}, \max \left\{ f(x_1), f(x_2), f\left(-\frac{b}{2a}\right) \right\} \right]$$

32. **Statement I :** If  $a + b = 0, a \neq 0$

then  $\sin^{-1} ax + \cos^{-1} bx = \frac{3\pi}{2}$  has no solution

**Statement II :**  $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, -1 \leq x \leq 1$

## INTEGER TYPE QUESTIONS

33. If  $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$  then the value of  $\frac{x^4 + y^4 + z^4 + 4x^2 y^2 z^2}{x^2 y^2 + y^2 z^2 + z^2 x^2}$  is

34. If the ordered pairs of natural numbers  $(a_1, b_1)$  and  $(a_2, b_2)$  are satisfying the equation  $\text{arc cot } 8 = \text{arc tan } a - \text{arc tan } b$ .

Then the value of  $\frac{a_1 + b_1 + a_2 + b_2}{9}$  is

35. If the sum of the series  $\cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \dots + \cot^{-1} 2n^2 + \dots$  upto  $\infty$  is  $\frac{k\pi}{4}$  then the value of  $K$  is

36. If  $\sum_{r=1}^{10} \tan^{-1} \left( \frac{3}{9r^2 + 3r - 1} \right) = \cot^{-1} \left( \frac{m}{n} \right)$  (where  $m$  and  $n$  are coprime) then the value of  $\frac{(2m+n)}{8}$  is

37. Let  $f(x) = (\text{arc tan } x)^3 + (\text{arc cot } x)^3$ . If the range of  $f(x)$  is  $[a, b]$  then the value of  $\frac{b}{7a}$  is

## JEE ADVANCED - VOL - II

38. If the area enclosed by the curves  $f(x) = \cos^{-1}(\cos x)$  and

$g(x) = \sin^{-1}(\cos x)$  in  $x \in \left[\frac{9\pi}{4}, \frac{15\pi}{4}\right]$  is

$a\pi^2 / b$  (where  $a$  and  $b$  are coprime), then the value of  $(a-b)$  is

39. Let  $\cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$ . If  $x$  satisfies the equation  $ax^3 + bx^2 + cx - 1 = 0$  then the value of  $(b-a-c)$  is

40. The number of solutions of the equation

$$|\tan^{-1} x| = \sqrt{(x^2 + 1)^2 - 4x^2} \text{ is}$$

## LEVEL-VI

### SINGLE CORRECT ANSWER TYPE

- |        |         |         |        |
|--------|---------|---------|--------|
| 1. (B) | 2. (C)  | 3. (A)  | 4. (B) |
| 5. (A) | 6. (A)  | 7. (A)  | 8. (B) |
| 9. (D) | 10. (D) | 11. (A) |        |

### MULTIPLE CORRECT ANSWER TYPE

- |              |           |
|--------------|-----------|
| 12. A,B,C, D | 13. A,B,C |
| 14. A,C,D    | 15. A,B,C |
| 16. A,D      | 17. B     |

### COMPREHENSION

- |       |       |       |       |
|-------|-------|-------|-------|
| 18. C | 19. B | 20. A | 21. A |
| 22. D | 23. B | 24. C |       |

### MATRIX MATCH TYPE

- |                               |
|-------------------------------|
| 25. A- Q, B-S, C-R, D-P       |
| 26. A- Q,R,S, B-Q, C-R,S, D-P |

### ASSERTION-REASONING TYPE

- |         |         |         |         |
|---------|---------|---------|---------|
| 28. (A) | 29. (A) | 30. (A) | 31. (A) |
| 32. (D) |         |         |         |

## INTEGER TYPE

- 33) 2      34) 8      35) 1      36) 4  
37) 4      38) 1      39) 2      40) 4

**LEVEL-VI HINTS****SINGLE CORRECT ANSWER**

- $\sin \cos^{-1}(\cos(\tan^{-1} x)) = p$   
 for  $x \in \mathbb{R}$ ,  $\tan^{-1} x \in (-\pi/2, \pi/2)$   
 $\cos^{-1}(\cos(\tan^{-1} x)) \in [0, \pi/2)$   
 $\sin(\cos^{-1}(\cos(\tan^{-1} x))) \in [0, 1)$
- $\sin^{-1} \sqrt{1-x^2} + \cos^{-1} x = \cot^{-1} \frac{\sqrt{1-x^2}}{x} - \sin^{-1} x$   
 $\frac{\pi}{2} + \sin^{-1} \sqrt{1-x^2} = \cot^{-1} \frac{\sqrt{1-x^2}}{x}$   
 $\tan^{-1} \frac{\sqrt{1-x^2}}{x} + \sin^{-1} \sqrt{1-x^2} = 0$   
 $\Rightarrow x \in [-1, 0) \cup \{1\}$
- $2 \cos^{-1} x = \cot^{-1} \left( \frac{2x^2 - 1}{2x\sqrt{1-x^2}} \right)$   
 Put  $x = \cos \theta$ :  $LHS = 2\theta$ ;  $0 \leq \theta \leq \pi$  and  $-1 \leq x \leq 1$  .....(i)  
 $RHS = \cot^{-1} \left( \frac{\cos 2\theta}{2 \cos \theta |\sin \theta|} \right) = \cot^{-1}(\cot 2\theta)$   
 $= 2\theta$  if  $2\theta < \pi$  .....(ii)  
 From Eqs (i) and (ii), we get  $0 < \theta < \pi/2$   
 $\therefore x \in (0, 1)$
- $f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{1+x^2} > 0 \forall x \in [-1, 1]$   
 $\therefore f(-1) \leq f(x) \leq f(1)$   
 $\Rightarrow -\frac{\pi}{2} + \frac{\pi}{4} \leq a \leq \frac{\pi}{2} - \frac{\pi}{4}$   
 $\Rightarrow -\frac{\pi}{4} \leq a \leq \frac{\pi}{4}$

**INVERSE TRIGONOMETRIC FUNCTIONS**

- $\cos^{-1}(\cos x) = x \forall x \in [0, \pi] \Rightarrow x = [x]$   
 $\Rightarrow x = 0, 1, 2, 3$   
 For  $x \in (\pi, 4)$  we get  $2\pi - x = [x]$   
 $\Rightarrow x = 2\pi - 3$
- $\frac{\pi}{2} - \cos^{-1} \cos \left( \frac{2(x^2 + 5|x| + 3) - 2}{x^2 + 5|x| + 3} \right)$   
 $= \cot \cot^{-1} \left( \frac{2}{9|x|} - 2 \right) + \frac{\pi}{2}$   
 $\frac{\pi}{2} - 2 + \frac{2}{x^2 + 5|x| + 3} = \frac{2}{9|x|} - 2 + \frac{\pi}{2}$   
 $\Rightarrow |x^2| - 4|x| + 3 = 0$   
 $|x| = 1, 3 \Rightarrow x = \pm 3, \pm 1$
- $1 \leq \sin^{-1} \cos^{-1} \sin^{-1} \tan^{-1} x \leq \frac{\pi}{2}$   
 $\Rightarrow \sin 1 \leq \cos^{-1} \sin^{-1} \tan^{-1} x \leq 1$   
 $\Rightarrow \cos \sin 1 \geq \sin^{-1}(\tan^{-1} x) \leq \cos 1$   
 $\Rightarrow \sin \cos \sin 1 \geq \tan^{-1} x \geq \sin \cos 1$   
 $\Rightarrow \tan \sin \cos \sin 1 \geq x \geq \tan \sin \cos 1$   
 $\therefore x \in [\tan \sin \cos 1, \tan \sin \cos \sin 1]$
- Let  $t = \frac{2x^2 + 4}{1 + x^2} \Rightarrow t \in (2, 4]$   
 $\Rightarrow \sin^{-1} \sin t < \pi - 3$   
 $\Rightarrow \sin^{-1} \sin(\pi - t) < \pi - 3 \Rightarrow t > 3$   
 $\Rightarrow 3 < t \leq 4 \Rightarrow |x| < 1$
- $2^{\pi/\sin^{-1} x} - 2(a+2)2^{\pi/\sin^{-1} x} + 8a < 0$   
 $(2^{\pi/\sin^{-1} x} - 4)(2^{\pi/\sin^{-1} x} - 2a) < 0$   
 Now  $2^{\pi/\sin^{-1} x} \in \left(0, \frac{1}{4}\right) \cup [4, \infty)$   
 Now for  $2^{\pi/\sin^{-1} x} \in \left(0, \frac{1}{4}\right)$

We have  $(2^{\pi/\sin^{-1}x} - 4) < 0$

$$\Rightarrow 2^{\pi/\sin^{-1}x} - 2a > 0$$

$$\Rightarrow 2a < 2^{\pi/\sin^{-1}x} \Rightarrow 2a < \frac{1}{4}$$

$$\Rightarrow 0 \leq a < \frac{1}{8}$$

Similarly, for  $2^{\pi/\sin^{-1}x} \in [4, \infty], a \leq 2$ , we get

$$a \in \left[0, \frac{1}{8}\right) \cup (2, \infty)$$

$$10. \left([\tan^{-1}x] - 4\right)^2 \leq 0 \Rightarrow [\tan^{-1}x] = 4$$

$$\Rightarrow 4 \leq \tan^{-1}x < 5 \Rightarrow x \in [\tan 4, \tan 5)$$

$$11. \frac{n^2 - 10n + 26}{2\sqrt{3}} < \sqrt{3} \text{ minimum value of } n = 3$$

### MULTIPLE CORRECT ANSWER

$$12. (a) y = \tan(\cos^{-1}x)$$

$$0 \leq \cos^{-1}x \leq \pi \text{ \& } \cos^{-1}x \neq \frac{\pi}{2}$$

$$\Rightarrow -1 \leq x \leq 1 \text{ \& } x \neq 0$$

$$(b) y = \tan(\cot^{-1}x)$$

$$0 < \cot^{-1}x < \pi \text{ \& } \cot^{-1}x \neq \frac{\pi}{2}$$

$$\therefore x \in R - \{0\}$$

$$(c) y = \sin(\tan^{-1}x)$$

$$\Rightarrow -\frac{\pi}{2} < \tan^{-1}x < \frac{\pi}{2} \Rightarrow x \in R$$

$$\therefore y \in (-1, 1)$$

$$(d) y = \cos(\tan^{-1}x) \Rightarrow y \in (0, 1] \text{ \& } x \in R$$

$$y = \frac{1}{\sqrt{1+x^2}} \text{ \& } y = \sin(\cot^{-1}x) \text{ has same graph}$$

$$13. \tan^{-1}y = 4 \tan^{-1}x = 2 \tan^{-1} \frac{2x}{1-x^2}$$

$$= \tan^{-1} \frac{\frac{4x}{1-x^2}}{1 - \frac{4x^2}{(1-x^2)^2}} = \tan^{-1} \frac{4x(1-x^2)}{x^4 - 6x^2 + 1}$$

$$\therefore y = \frac{4x(1-x^2)}{x^4 - 6x^2 + 1}$$

$$y = \infty \Rightarrow x^4 - 6x^2 + 1 = 0$$

$$\text{Also } \tan^{-1}y = 4 \tan^{-1}x = \frac{\pi}{2}$$

$$14. x^2 + x + 1 = \lambda x + 1 \text{ \& } -1 \leq x^2 + x + 1 \leq 1$$

$$\Rightarrow x = 0 \text{ or } \lambda = 1 \quad \therefore x = 0 \text{ is one solution}$$

$$\text{\& } -1 \leq \lambda - 1 < 0 \Rightarrow 0 \leq \lambda < 1$$

$$\therefore x = 0$$

$$15. \sin\left(2 \cos^{-1}\left\{\cot\left(2 \tan^{-1}x\right)\right\}\right) = 0$$

$$\Rightarrow 2 \cos^{-1}\{\cot(2 \tan^{-1}x)\} = n\pi, n \in \mathbb{I}$$

$$\Rightarrow \cos^{-1}(\cot\{2 \tan^{-1}x\}) = \frac{n\pi}{2} = 0, \frac{\pi}{2}, \pi$$

$$(\because 0 \leq \cos^{-1}x \leq \pi)$$

$$\Rightarrow \cot\left\{\tan^{-1}\left(\frac{2x}{1-x^2}\right)\right\} = 1, 0, -1$$

$$\Rightarrow \cot\left\{\cot^{-1}\left(\frac{1-x^2}{2x}\right)\right\} = 1, 0, -1$$

$$\Rightarrow \frac{1-x^2}{2x} = 1, 0, -1$$

$$\Rightarrow -x^2 = 2x, 0, -2x.$$

$$\text{or } x^2 + 2x - 1 = 0, x^2 - 1 = 0, x^2 - 2x - 1 = 0$$

$$\text{or } x = 1, -1 \quad x = -1 \pm \sqrt{2}, x = \pm 1, x = 1 \pm \sqrt{2}$$

$$16. \text{ For the given equation } 0 \leq x, y \leq 1$$

$$\text{Also, } \sin^{-1}x + \sin^{-1}y = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}x = \cos^{-1}y = \sin^{-1}\sqrt{1-y^2}$$

$$\Rightarrow x = \sqrt{1-y^2} \Rightarrow x^2 + y^2 = 1$$

Again,  $\sin 2x = \cos 2y$

$$\Rightarrow \cos\left(\frac{\pi}{2} - 2x\right) = \cos 2y$$

$$\Rightarrow \frac{\pi}{2} - 2x = 2n\pi \pm 2y, \text{ where } n \in I$$

$$\Rightarrow x \pm y = \frac{\pi}{4} - n\pi$$

From equations (1) and (2), we get

$$x = \frac{\pi}{8} + \sqrt{\frac{1}{2} - \frac{\pi^2}{64}} \text{ and}$$

$$y = \sqrt{\frac{1}{2} - \frac{\pi^2}{64}} - \frac{\pi}{8}$$

17.  $f(x) = \sin^{-1}|\sin kx| + \cos^{-1}(\cos kx)$

Let  $g(x) = \sin^{-1}|\sin x| + \cos^{-1}(\cos x)$

$$g(x) = \begin{cases} 2x, & 0 \leq x \leq \frac{\pi}{2} \\ \pi, & \frac{\pi}{2} < x \leq \frac{3\pi}{2} \\ 4\pi - 2x, & \frac{3\pi}{2} < x \leq 2\pi \end{cases}$$

$g(x)$  is periodic with period  $2\pi$  and is constant in the continuous interval

$$\left[2n\pi + \frac{\pi}{2}, 2n\pi + \frac{3\pi}{2}\right]$$

(Where  $n \in I$  and  $f(x) = g(kx)$ )

So,  $f(x)$  is constant in the interval

$$\left[\frac{2n\pi}{k} + \frac{\pi}{2k}, \frac{2n\pi}{k} + \frac{3\pi}{2k}\right]$$

$$\Rightarrow \frac{\pi}{4} = \frac{3\pi}{2k} - \frac{\pi}{2k} \Rightarrow \frac{\pi}{k} = \frac{\pi}{4} \Rightarrow k = 4$$

### COMPREHENSION

18. Let  $\sin^{-1}(-x) = \theta$

$$x = -\sin \theta, \quad \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2} = \sin(2\pi - \theta)$$

$$2\pi - \theta = \sin^{-1} x$$

$$2\pi - \sin^{-1} x = \theta = \sin^{-1}(-x)$$

19. Let  $\cos^{-1}(-x) = \theta, \pi \leq \theta \leq 2\pi$

$$\therefore -x = \cos \theta$$

$$x = -\cos \theta = \cos(3\pi - \theta)$$

$$\therefore 3\pi - \theta = \cos^{-1} x$$

$$\therefore 3\pi - \cos^{-1} x = \theta = \cos^{-1}(-x)$$

Calculate  $f(-x)$ .

20.  $f = (\sin^{-1} x)^3 - (\cos^{-1} x)^3, -1 \leq x \leq 1$

$$f' = 3(\sin^{-1} x)^2 \frac{1}{\sqrt{1-x^2}} - 3(\cos^{-1} x)^2 \cdot \left(\frac{-1}{\sqrt{1-x^2}}\right)$$

$$f' < 0$$

$$\therefore \text{Minimum value} = \left(\frac{\pi}{2}\right)^3 - (2\pi)^3$$

$$= \frac{\pi^3}{8} - 8\pi^3 = \frac{-63\pi^3}{8}$$

21.  $\frac{-5\pi}{2} \leq \sin^{-1} x \leq \frac{-3\pi}{2}$

$$\therefore \frac{3\pi}{2} \leq -\sin^{-1} x \leq \frac{5\pi}{2}$$

$$\frac{3\pi}{2} + \frac{5\pi}{2} \leq \frac{5\pi}{2} - \sin^{-1} x \leq \frac{5\pi}{2} + \frac{5\pi}{2}$$

$$4\pi \leq \cos^{-1} x \leq 5\pi$$

22 to 24.

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\cos^{-1} y \in [0, \pi]$$

$$\sec^{-1} z \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$

## INVERSE TRIGONOMETRIC FUNCTIONS

$$\Rightarrow \sin^{-1} x + \cos^{-1} y + \sec^{-1} z \leq \frac{\pi}{2} + \pi + \pi = \frac{5\pi}{2}$$

$$\text{Also } t^2 - \sqrt{2\pi} t + 3\pi$$

$$= t^2 - 2\sqrt{\frac{\pi}{2}} t + \frac{\pi}{2} - \frac{\pi}{2} + 3\pi = \left(t - \sqrt{\frac{\pi}{2}}\right)^2 + \frac{5\pi}{2} \geq \frac{5\pi}{2}$$

The given inequality holds

$$\Leftrightarrow x = 1, y = -1, z = -1$$

$$\text{LHS} = \text{RHS} = \frac{5\pi}{2}$$

$$\Rightarrow x = 1, y = -1, z = -1 \text{ and}$$

$$t = \sqrt{\frac{\pi}{2}} \Rightarrow \cos^{-1}(\cos 5t^2) = \cos^{-1}\left(\cos\left(\frac{5\pi}{2}\right)\right) = \frac{\pi}{2}$$

$$\cos^{-1}(\min\{x, y, z\}) = \cos^{-1}(-1) = \pi$$

### MATRIX MATCH TYPE

25. (i) When  $x \in (0, \cos 1)$ , then  $\sin^{-1} x \in (0, \pi/2 - 1)$   
and  $\cos^{-1} x \in (1, \pi/2)$   
(ii) When  $x \in (\cos 1, 1/\sqrt{2})$ , then  
 $\sin^{-1} x \in (\pi/2 - 1, \pi/4)$  and  $\cos^{-1} x \in (\pi/4, 1)$   
(iii) when  $x \in (1/\sqrt{2}, \sin 1)$ , then  
 $\sin^{-1} x \in (1/\sqrt{2}, 1)$  and  $\cos^{-1} x \in (\pi/2 - 1, \pi/4)$   
(iv) When  $x \in (\sin 1, 1)$ , then  $\sin^{-1} x \in (1, \pi/2)$   
and  $\cos^{-1} x \in (0, \pi/2 - 1)$

26. a)  $(\sin^{-1} x)^2 + (\sin^{-1} y)^2 = \frac{\pi^2}{2}$

$$\Rightarrow (\sin^{-1} x)^2 = (\sin^{-1} y)^2 = \frac{\pi^2}{4}$$

$$\Rightarrow \sin^{-1} x = \pm \frac{\pi}{2}, \sin^{-1} y = \pm \frac{\pi}{2}$$

$$\Rightarrow x = \pm 1 \text{ and } y = \pm 1$$

$$\Rightarrow x^3 + y^3 = -2, 0, 2$$

b)  $(\cos^{-1} x)^2 + (\cos^{-1} y)^2 = 2\pi^2$

$$\Rightarrow (\cos^{-1} x)^2 = (\cos^{-1} y)^2 = \pi$$

## JEE ADVANCED - VOL - II

$$\Rightarrow x = y = -1$$

$$\Rightarrow x^5 + y^5 = -2$$

c)  $(\sin^{-1} x)^2 (\cos^{-1} y)^2 = \frac{\pi^4}{4}$

$$\Rightarrow (\sin^{-1} x)^2 = \frac{\pi^2}{4} \text{ and } (\cos^{-1} y)^2 = \pi^2$$

$$\Rightarrow (\sin^{-1} x) = \pm \frac{\pi}{2} \text{ and } (\cos^{-1} y) = \pi$$

$$\Rightarrow x = \pm 1 \text{ and } y = -1$$

$$\Rightarrow |x - y| = 0, 2$$

27. (A) If  $a = 1, b = 0$ , then  $\sin^{-1} x + \cos^{-1} y = 0$

$$\Rightarrow \sin^{-1} x = -\cos^{-1} y \Rightarrow x^2 + y^2 = 1$$

- (B) If  $a = 1, b = 1$ , then

$$\sin^{-1} x + \cos^{-1} y + \cos^{-1} xy = \frac{\pi}{2}$$

$$\Rightarrow \cos^{-1} x - \cos^{-1} y = \cos^{-1} xy$$

$$\Rightarrow xy + \sqrt{1-x^2}\sqrt{1-y^2} = xy$$

$$\Rightarrow (x^2 - 1)(y^2 - 1) = 0$$

(C) For  $a = 1$  &  $b = 2$ ,  $(1 - x^2)(1 - y^2) = x^2 y^2$

(D) For  $a = 2, b = 2$ ,  $(1 - y^2)(1 - 4x^2) = 0$

### ASSERTION-REASONING TYPE

28.  $x = \tan \theta, \quad -\frac{\pi}{2} < \theta < 0$

$$\sec^{-1} \sqrt{1+x^2} = \sec^{-1} |\sec \theta|$$

$$= \sec^{-1} \sec \theta = -\theta = -\tan^{-1} x$$

$$\cos^{-1} \frac{1-x^2}{1+x^2} = \cos^{-1} \cos 2\theta$$

$$= -2\theta = -2 \tan^{-1} x$$

29. We know that

$$\sin^{-1}(\sin \theta) = \theta, \forall \theta \in [-\pi/2, \pi/2]$$

$$\sin^{-1} \sin(5 - 2\pi) = 5 - 2\pi = 5 - 2 \times 3.14$$

$$= -1.28 \in [-\pi/2, \pi/2]$$

$$\sin^{-1} \sin 5 > x^2 - 4x \Rightarrow x^2 - 4x + 2\pi - 5 < 0$$

$$\Rightarrow (x-2)^2 - (\sqrt{9-2\pi})^2 < 0$$

$$\Rightarrow -\sqrt{9-2\pi} < x-2 < \sqrt{9-2\pi}$$

$$\Rightarrow 2-\sqrt{9-2\pi} < x < 2+\sqrt{9-2\pi}$$

$\Rightarrow$  inequality holds for number of values.

$\Rightarrow$  Assertion (A) and reason (R) both are true and reason (R) is the correct explanation of the assertion (A)

30. Both I and II are true and II is the correct explanation of I

$$31. f(x) = \left( \cos^{-1} x - \frac{\pi}{4} \right)^2 + \frac{3\pi^2}{16}$$

$$-\frac{\pi}{4} \leq \cos^{-1} x - \frac{\pi}{4} \leq \frac{3\pi}{4}$$

$$\Rightarrow 0 \leq \left( \cos^{-1} x - \frac{\pi}{4} \right)^2 \leq \frac{9\pi^2}{16}$$

$$f(x) = \left( \cos^{-1} x \right)^2 - \frac{\pi}{2} \cos^{-1} x + \frac{\pi^2}{4}$$

$$g(t) = t^2 - \frac{\pi}{2}t + \frac{\pi^2}{4} \quad t \in [0, \pi]$$

$$\min g(t) = \min \left\{ g(0), g(\pi), g\left(\frac{\pi}{4}\right) \right\} = \frac{3\pi^2}{16}$$

$$\max g(t) = \max \left\{ g(0), g(\pi), g\left(\frac{\pi}{4}\right) \right\} = \frac{3\pi^2}{4}$$

32. Given equation has a solution when  $ax=1$

### INTEGER TYPE

$$33. x = y = z = \frac{\sqrt{3}}{2}$$

$$34. \text{ Given that } \tan^{-1}\left(\frac{1}{8}\right) = \tan^{-1}a - \tan^{-1}b$$

$$= \tan^{-1}\left(\frac{a-b}{1+ab}\right) \Rightarrow \frac{1}{8} = \frac{a-b}{1+ab}$$

$$\Rightarrow 1+ab = 8a-8b$$

$$\Rightarrow b(a+8) = 8a-1 \Rightarrow b = \frac{8a-1}{a+8}$$

(Note  $a \geq 1$ )

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$$\therefore b = \frac{8(a+8)-65}{a+8} = 8 - \frac{65}{a+8} = 8 - \frac{(5)(13)}{(a+8)}$$

### INVERSE TRIGONOMETRIC FUNCTIONS

Hence  $a = 5$  or  $57$

(In order that  $b$  may be an integer.)

$$\text{If } a = 5 \Rightarrow b = 3$$

$$\text{If } a = 57 \Rightarrow b = 7$$

$$\text{Hence } \frac{a_1 + b_1 + a_2 + b_2}{9} = 8$$

35. Let us denote the  $n^{\text{th}}$  terms of the series by  $t_n$  then we have

$$t_n = \cot^{-1} 2n^2 = \tan^{-1} \frac{1}{2n^2}$$

$$= \tan^{-1} \left( \frac{2}{4n^2} \right) = \tan^{-1} \left( \frac{2}{4n^2 - 1 + 1} \right)$$

$$= \tan^{-1} \left[ \frac{(2n+1) - (2n-1)}{(2n-1)(2n+1) + 1} \right]$$

$$= \tan^{-1} (2n+1) - \tan^{-1} (2n-1) \text{ or}$$

$$t_n = \tan^{-1} (2n+1) - \tan^{-1} (2n-1)$$

Putting  $n = 1, 2, 3, \dots$  etc. in the above equation we have each term as

$$t_1 = \tan^{-1} 3 - \tan^{-1} 1$$

$$t_2 = \tan^{-1} 5 - \tan^{-1} 3$$

$$t_3 = \tan^{-1} 7 - \tan^{-1} 5$$

$$\dots\dots\dots$$

$$\dots\dots\dots$$

$$t_n = \tan^{-1} (2n+1) - \tan^{-1} (2n-1)$$

$$\dots\dots\dots$$

$$\text{adding, } S_n = \tan^{-1} (2n+1) - \tan^{-1} 1$$

$$\text{as } n \rightarrow \infty, 2n+1 \rightarrow \infty \text{ and } \tan^{-1} 1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$K = 1.$$

$$36. \text{ We have } a_n = \tan^{-1} \left( \frac{3}{9n^2 + 3n - 1} \right)$$

$$= \tan^{-1} \left( \frac{3}{1 + (3n+2)(3n-1)} \right)$$

$$= \tan^{-1} (3n+2) - \tan^{-1} (3n-1)$$

$$\therefore \text{ Sum of first 10 terms} = \sum_{r=1}^{10} a_r$$

$$= \sum_{r=1}^{10} \left( \tan^{-1} (3r+2) - \tan^{-1} (3r-1) \right)$$

$$= (\tan^{-1} 5 - \tan^{-1} 2) + (\tan^{-1} 8 - \tan^{-1} 5) + \dots\dots\dots + (\tan^{-1} 32 - \tan^{-1} 29)$$



$$= \tan^{-1} 32 - \tan^{-1} 2 = \tan^{-1} \left( \frac{32-2}{1+32 \cdot 2} \right)$$

$$= \tan^{-1} \frac{30}{65} = \tan^{-1} \left( \frac{6}{13} \right) = \cot^{-1} \left( \frac{13}{6} \right)$$

$$= \cot^{-1} \left( \frac{m}{n} \right)$$

$$\therefore m = 13 \text{ and } n = 6.$$

$$\text{Hence } \frac{(2m+n)}{8} = 4$$

$$\begin{aligned} 37. \text{ We have } f(x) &= (\tan^{-1} x)^3 + (\cot^{-1} x)^3 \\ &= (\tan^{-1} x + \cot^{-1} x) \\ &\quad \left( (\tan^{-1} x)^2 - (\tan^{-1} x)(\cot^{-1} x) + (\cot^{-1} x)^2 \right) \\ &= \frac{\pi}{2} \left( (\tan^{-1} x)^2 - (\tan^{-1} x) \left( \frac{\pi}{2} - \tan^{-1} x \right) + \left( \frac{\pi}{2} - \tan^{-1} x \right)^2 \right) \end{aligned}$$

$$\left( \text{Using } \cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x \right)$$

$$= \frac{3\pi}{2} \left( \left( \tan^{-1} x - \frac{\pi}{4} \right)^2 + \frac{\pi^2}{48} \right)$$

Clearly,  $f(x)$  will be minimum when

$$\left( \tan^{-1} x - \frac{\pi}{4} \right)^2 = 0$$

and  $f(x)$  will be maximum when

$$\left( \tan^{-1} x - \frac{\pi}{4} \right)^2 = \left( -\frac{\pi}{2} - \frac{\pi}{4} \right)^2$$

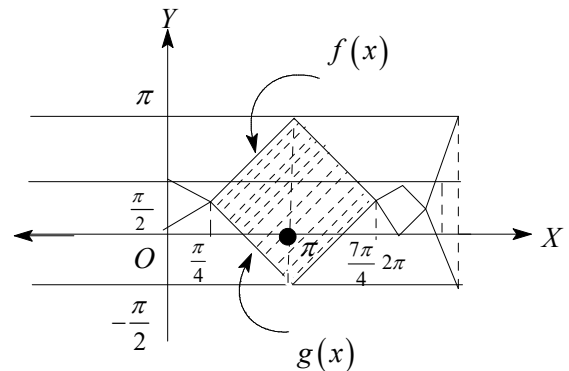
$$\therefore a = f(x)_{\min} = \frac{3\pi}{2} \left( 0 + \frac{\pi^2}{48} \right) = \frac{\pi^3}{32} \text{ and } b = f$$

$$(x)_{\max} = \frac{3\pi}{2} \left( \left( \frac{-3\pi}{4} \right)^2 + \frac{\pi^2}{48} \right) = \frac{7\pi^3}{8}$$

$$\text{Hence } \frac{b}{7a} = \frac{\frac{\pi^3}{8}}{\frac{\pi^3}{32}} = 4$$

38. We have

$$g(x) = \sin^{-1}(\cos x) = \frac{\pi}{2} - \cos^{-1}(\cos x)$$



Both the curves bound the regions of same area

in  $\left[ \frac{\pi}{4}, \frac{7\pi}{4} \right], \left[ \frac{9\pi}{4}, \frac{15\pi}{4} \right]$  and so on

Therefore, the required area = area of shaded

$$\text{square} = \frac{9\pi^2}{8} = \frac{a\pi^2}{b}$$

$$\therefore a = 9 \text{ and } b = 8 \Rightarrow a - b = 1$$

$$39. \cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$$

$$\Rightarrow \cos^{-1}(2x) + \cos^{-1}(3x)$$

$$= \pi - \cos^{-1}(x) = \cos^{-1}(-x)$$

$$\Rightarrow \cos^{-1} \left[ (2x)(3x) - \sqrt{1-4x^2} \sqrt{1-9x^2} \right]$$

$$= \cos^{-1}(-x)$$

$$\Rightarrow 6x^2 - \sqrt{1-4x^2} \sqrt{1-9x^2} = -x$$

$$\Rightarrow (6x^2 + x)^2 = (1-4x^2)(1-9x^2)$$

$$\Rightarrow x^2 + 12x^3 = 1 - 13x^2$$

$$\Rightarrow 12x^3 + 14x^2 - 1 = 0$$

$$\Rightarrow a = 12; b = 14; c = 1$$

$$\Rightarrow b - a - c = 14 - 12 + 1 = 3$$

$$40. |\tan^{-1} x| = |x^2 - 1|$$

From the graph we get number of solutions is 4.